



如何控制、校正超導量子位元？(下) & 超導量子電腦硬體元件導論

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2026.05.11



Key Information

- 超導量子計算(Superconducting Quantum Computation, SC QC)，是在極低溫下，以超導(Superconductor)製作的約瑟夫森節(Josephson Junction)為基礎，設計出非線性的LC電路作為量子位元 (Qubit)，搭配電路設計做出可控制、可讀出、可計算的量子處理器(QPU)。整機運作結合微波工程、低溫物理、材料、電路設計與量子控制。
- 04/20：超導量子計算是什麼？
- 04/27：超導量子位元的特性、超導量子電路
- 05/04：如何控制、校正超導量子位元？
- 05/11：超導量子電腦硬體元件導論

Quantum Phase

$$i\hbar\partial_t|\psi\rangle = \hat{H}_0|\psi\rangle \quad |\psi\rangle = \sum_i c^i |\psi_i\rangle \quad \hat{H}_0|\psi_i\rangle = E^i|\psi_i\rangle$$

$$|\psi_i(t)\rangle = \exp\left(-i\frac{E^i}{\hbar}t\right) * |\psi_i(0)\rangle$$

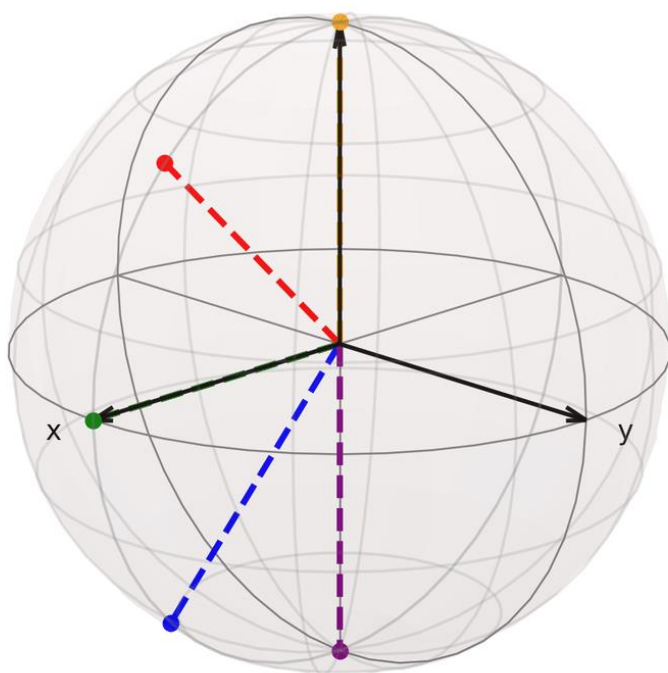
Phase

$$|\psi(t)\rangle = \left[\exp\left(-\frac{i}{\hbar}T\left[\int_0^t \hat{H}_0(s)ds\right]\right) \right] |\psi(0)\rangle$$

$U(t, 0)$

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{\Delta t} |+\rangle = \frac{e^{-i\frac{E^0}{\hbar}t}|0\rangle + e^{-i\frac{E^1}{\hbar}t}|1\rangle}{\sqrt{2}}$$

Lab frame
z



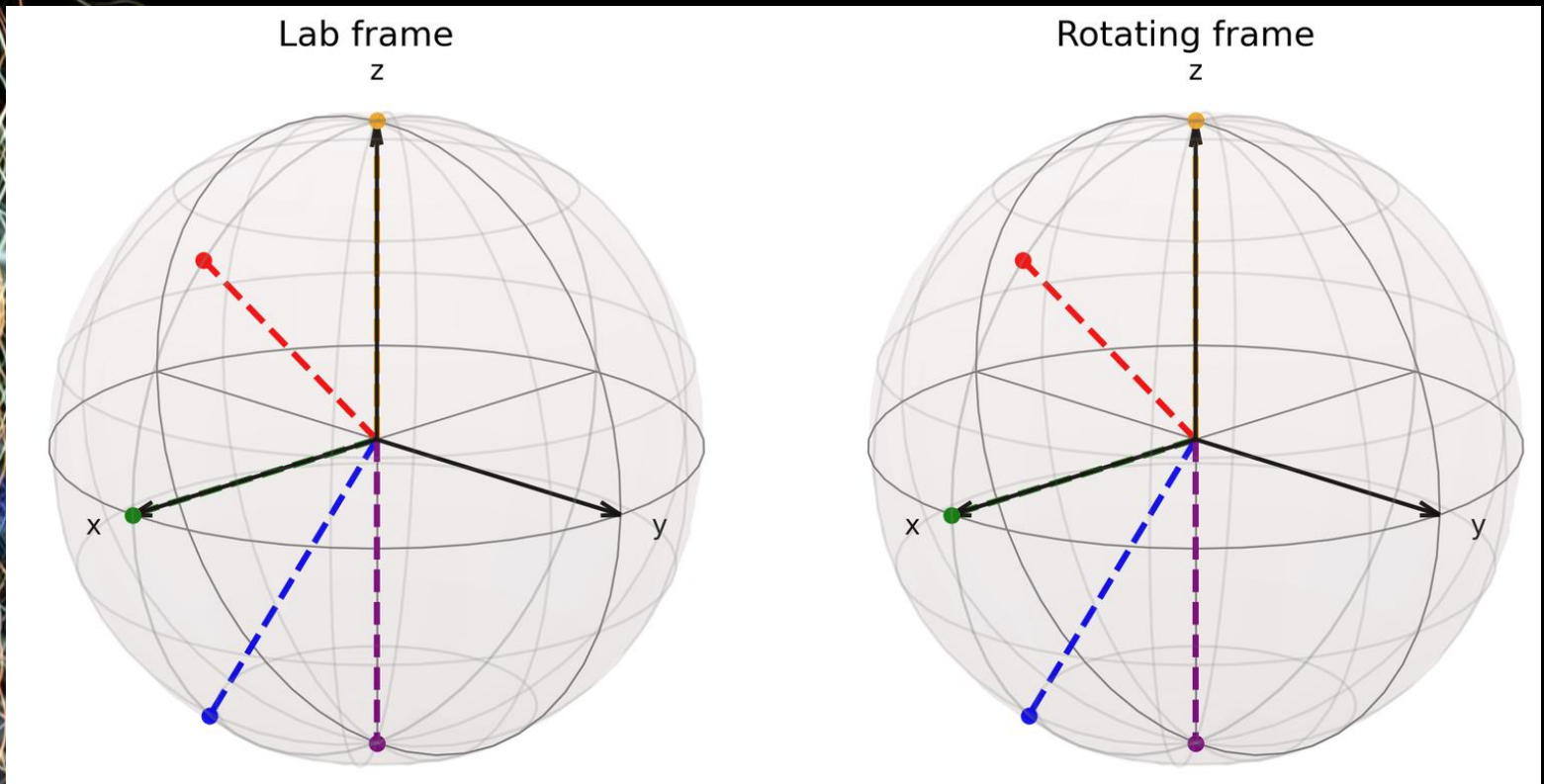
Rotating Frame

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}_0 |\psi(t)\rangle$$

$$|\Theta(t)\rangle = U^\dagger(t, 0) |\psi(t)\rangle$$

$$i\hbar \partial_t |\Theta\rangle = \hat{H}_r |\Theta\rangle = 0$$

$$U^\dagger(t, 0) = \left[\exp \left(\frac{i}{\hbar} T \left[\int_0^t \hat{H}_0(s) ds \right] \right) \right] \quad \hat{H}_r = 0$$



Interaction Basis

$$\hat{H} = \underbrace{\hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right)}_{\hat{H}_0} + \underbrace{\hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})}_{\hat{V}(t)}$$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle = [\hat{H}_0 + \hat{V}(t)] |\psi(t)\rangle$$

$$|\Theta(t)\rangle \equiv U_0^{\dagger}(t, 0) |\psi(t)\rangle$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = i\hbar \frac{\partial}{\partial t} [U_0^{\dagger}(t, 0) |\psi(t)\rangle]$$

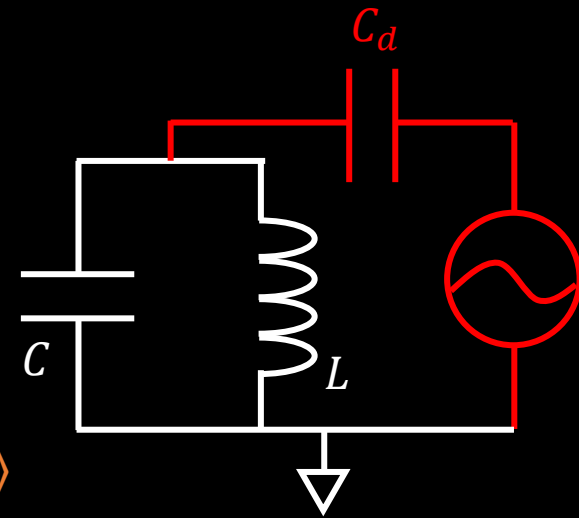
$$= \left[i\hbar \frac{\partial}{\partial t} U_0^{\dagger}(t, 0) \right] |\psi(t)\rangle + U_0^{\dagger}(t, 0) * i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$$

$$= -U_0^{\dagger}(t, 0) \hat{H}_0 |\psi(t)\rangle + U_0^{\dagger}(t, 0) * [\hat{H}_0 + \hat{V}(t)] |\psi(t)\rangle$$

$$= U_0^{\dagger}(t, 0) * \hat{V}(t) |\psi(t)\rangle$$

$$= \underbrace{U_0^{\dagger}(t, 0) * \hat{V}(t) * U_0(t, 0)}_{\hat{H}_r} * \underbrace{U_0^{\dagger}(t, 0) |\psi(t)\rangle}_{|\Theta(t)\rangle}$$

$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = \hat{H}_r |\Theta(t)\rangle$$



$$i\hbar \frac{\partial}{\partial t} U_0(t, 0) = \hat{H}_0 U_0(t, 0)$$

$$\rightarrow -i\hbar \frac{\partial}{\partial t} U_0^{\dagger}(t, 0) = U_0^{\dagger}(t, 0) \hat{H}$$

Driving Frequency

$$\hat{H} = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

$$\hat{H}_r = U_0^{\dagger} * [\hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})] * U_0$$

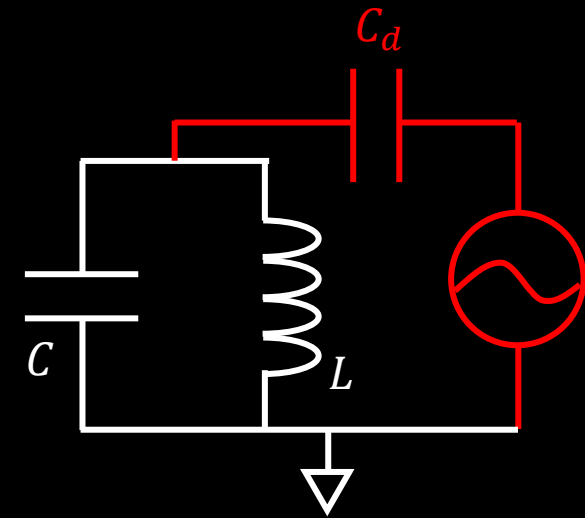
$$= \hbar\Omega * f(t) * i(e^{i\omega_{\Sigma}t} \hat{a}^{\dagger} - e^{-i\omega_{\Sigma}t} \hat{a})$$

$$= \hbar\Omega * f(t) * i[(\cos \omega_{\Sigma}t + i \sin \omega_{\Sigma}t) \hat{a}^{\dagger} - c.c.]$$

$$f(t) = \sin(\omega_{\Sigma}t)$$

$$= \hbar\Omega * [\cos(0) \hat{a}^{\dagger} + \cos(2\omega_{\Sigma}t) \hat{a}^{\dagger} + \dots - c.c.]$$

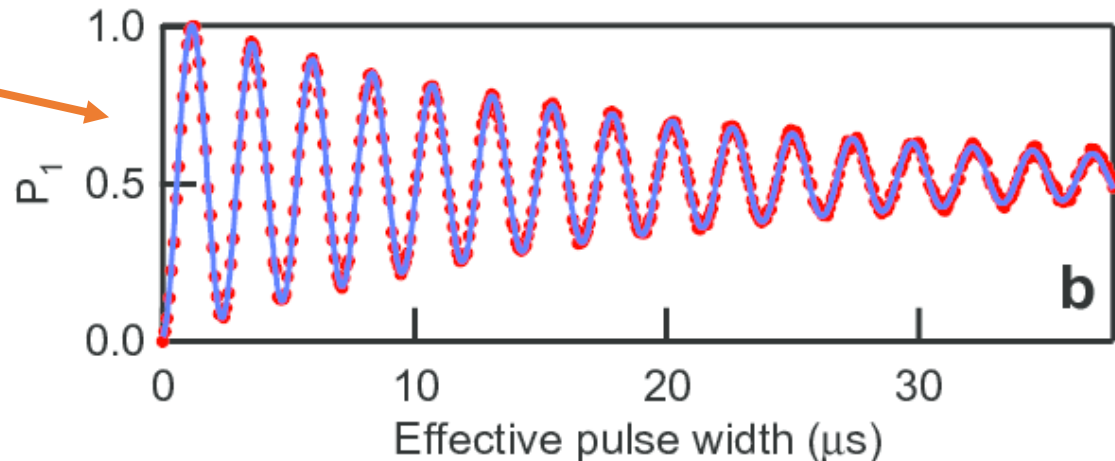
$$= \hbar\Omega * (\hat{a}^{\dagger} - \hat{a}) + \cos(2\omega_{\Sigma}t) [...]$$



$$V_s(t) = V_0 * f(t)$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

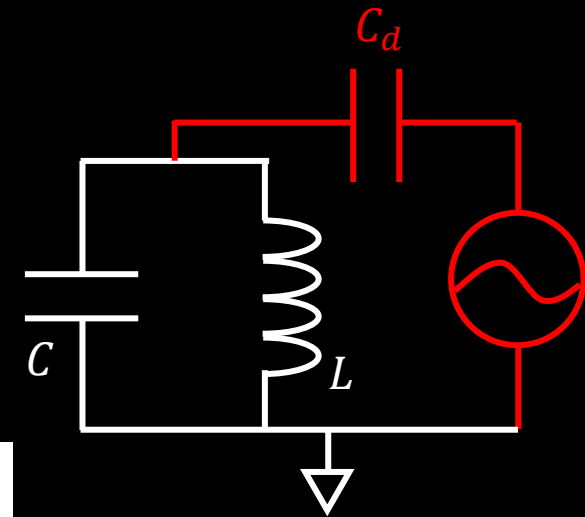
$$\sin A \cos B = \frac{1}{2} [\sin(A - B) - \sin(A + B)]$$



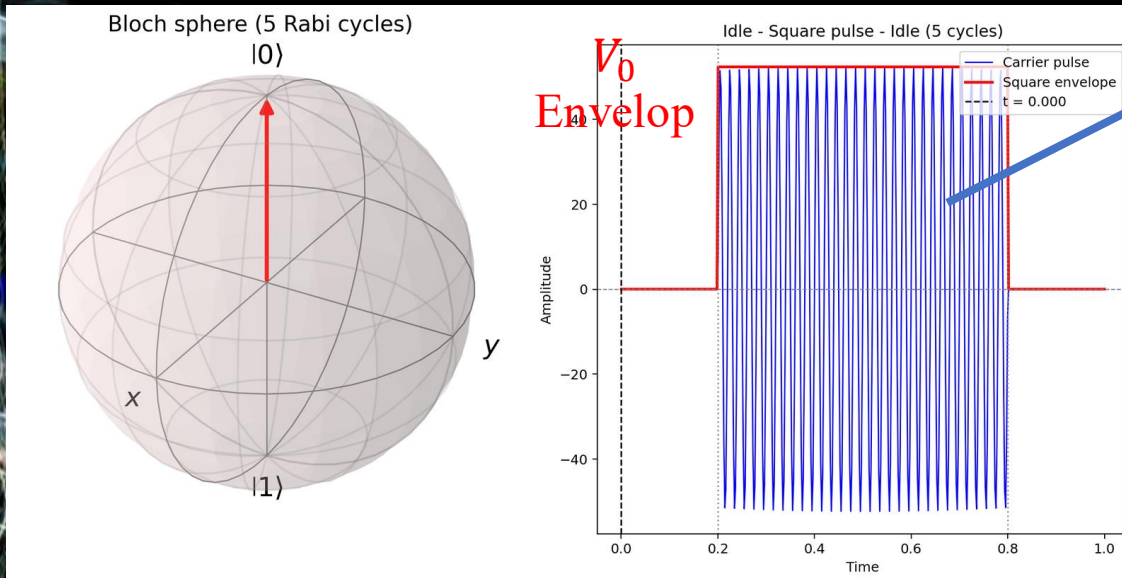
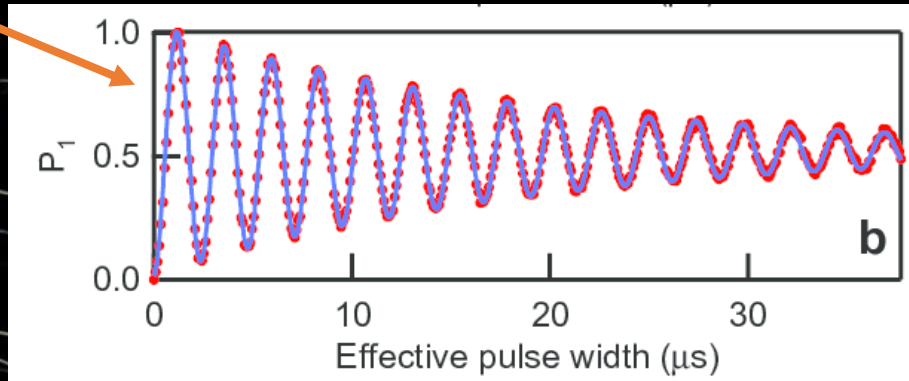
Driving Frequency

$$\hat{H} = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

$$\hat{H}_r = \hbar\Omega * (\hat{a}^{\dagger} - \hat{a}) + \dots$$



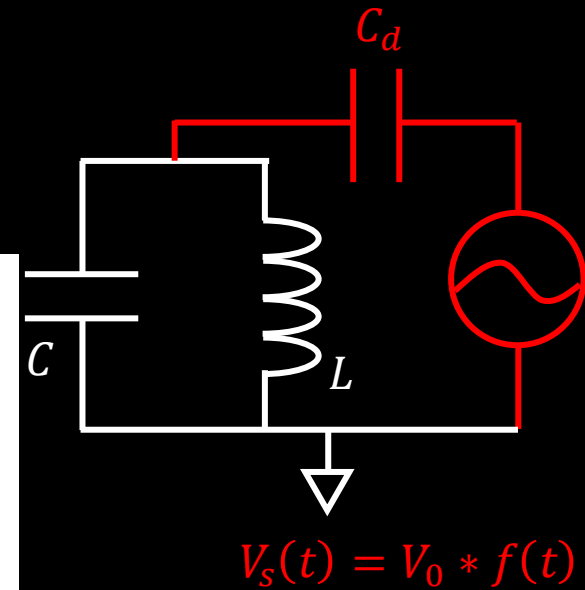
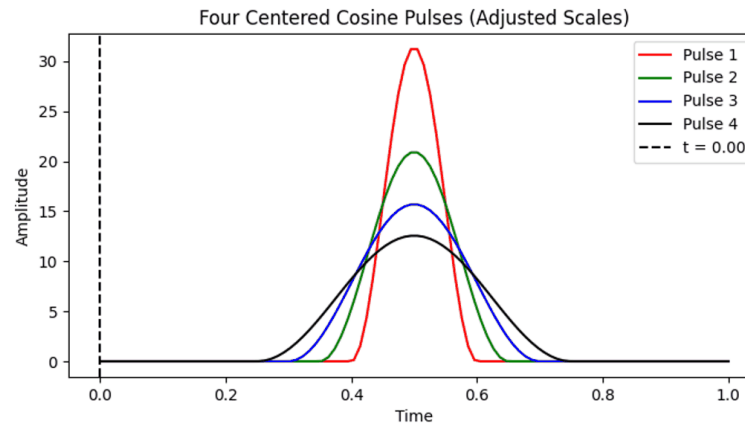
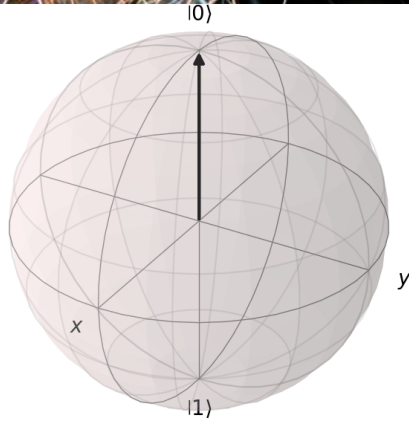
$$V_s(t) = V_0 * \sin \omega t$$



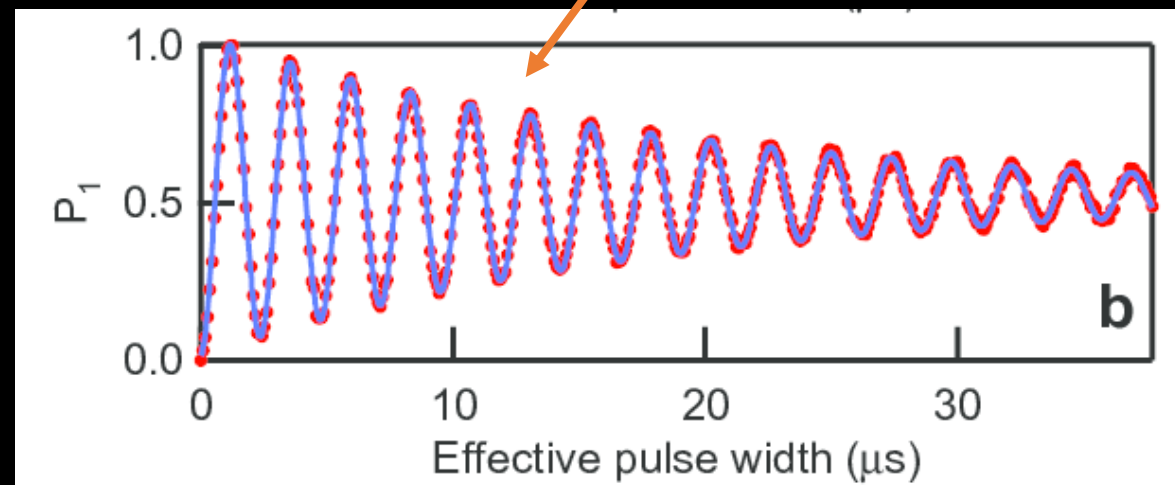
$$f(t) = \sin(\omega_{\Sigma} t)$$

Carrier Frequency

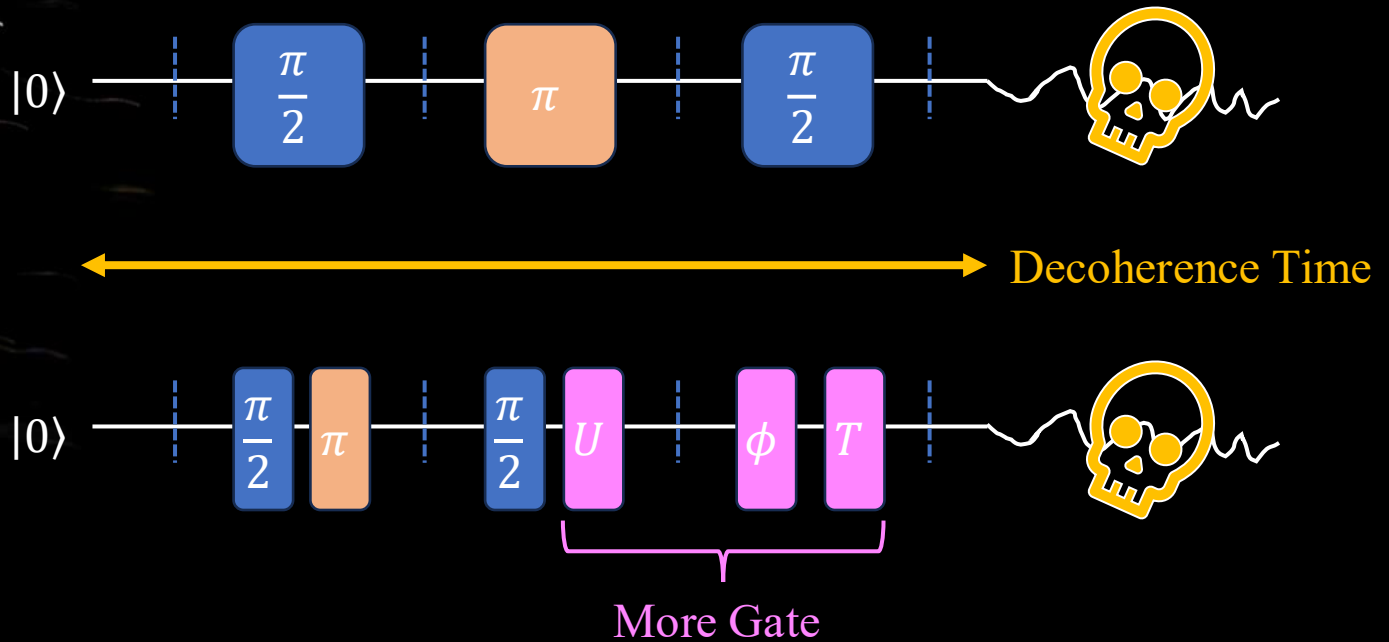
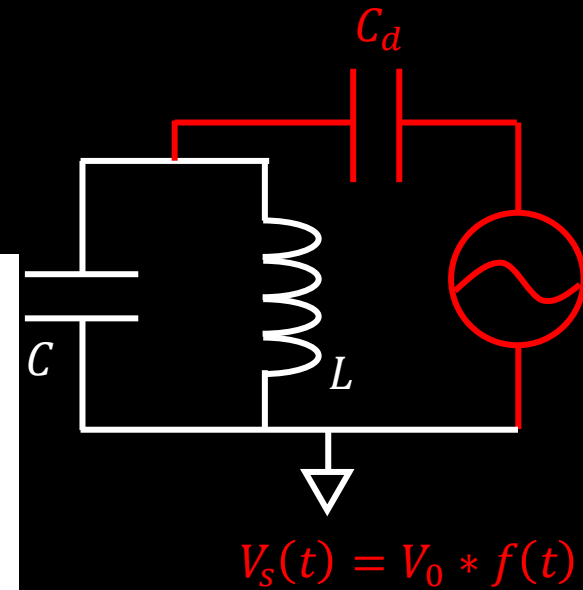
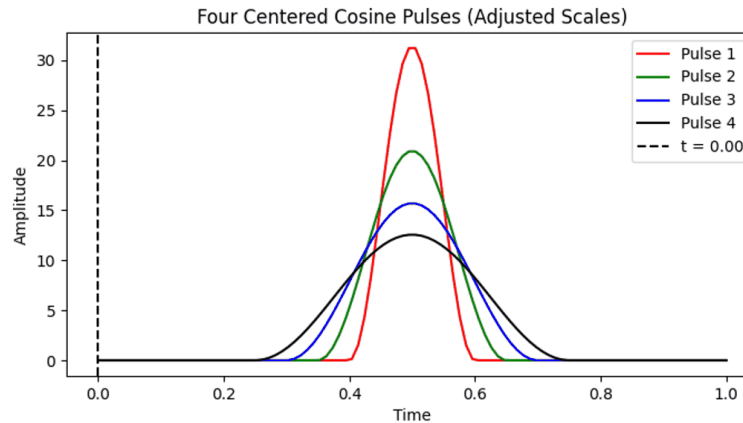
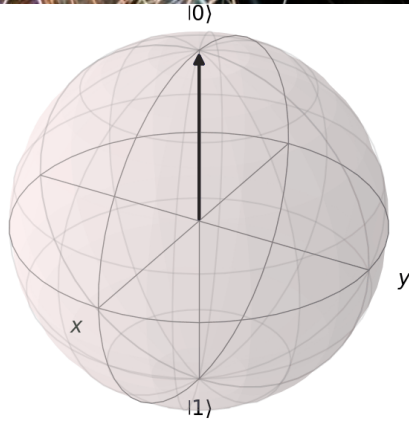
Pulse Calibration



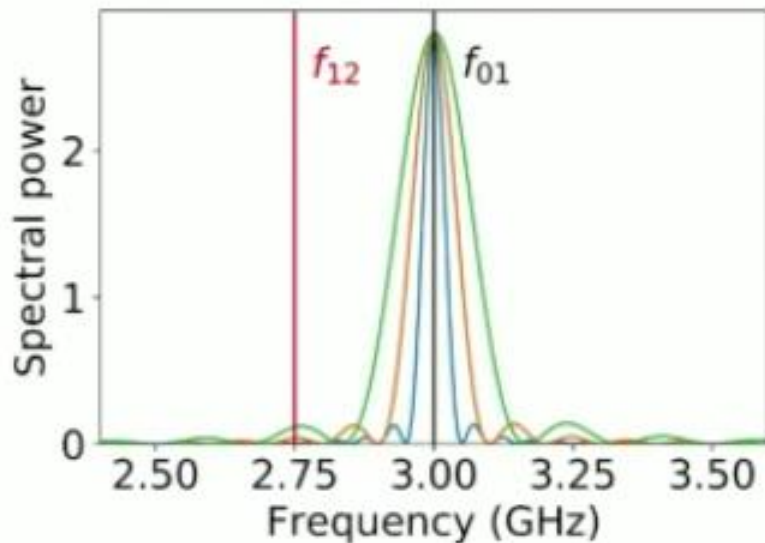
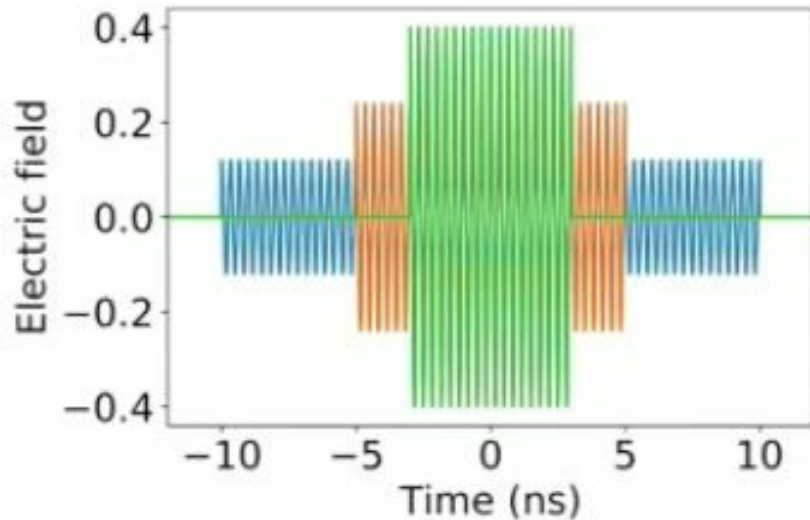
$$\text{Rabi Frequency } \Omega = C_d V_0 \sqrt{\frac{\omega_\Sigma}{2\hbar(C + C_d)}}$$



Pulse Calibration

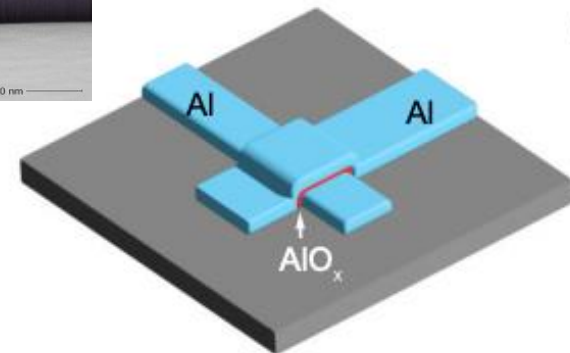
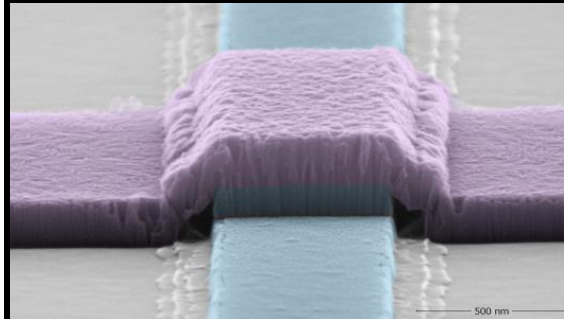


Pulse Calibration



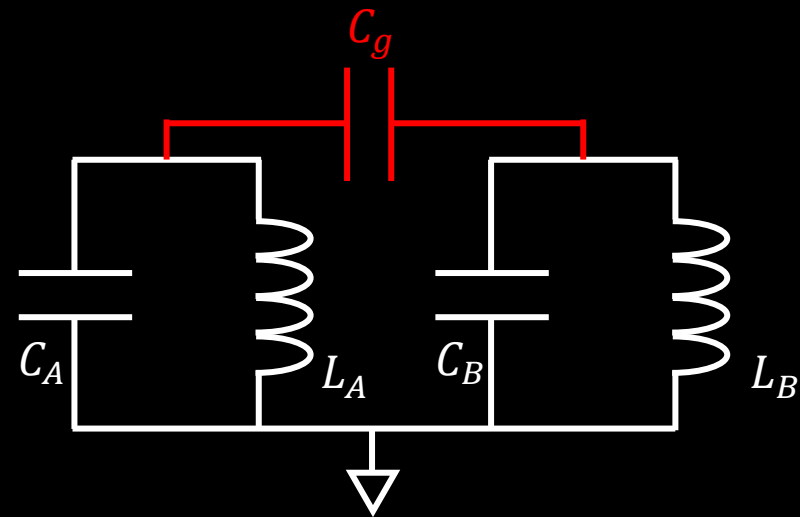
Narrow Pulse

- More Fourier Component
 - $|1\rangle \rightarrow |2\rangle$
 - Higher Frequency Photon
 - Quasi-particle excitation
 - More energy waste $\rightarrow$ Heating
- Junction Change, Qubit Frequency Change, or Qubit Destroy.



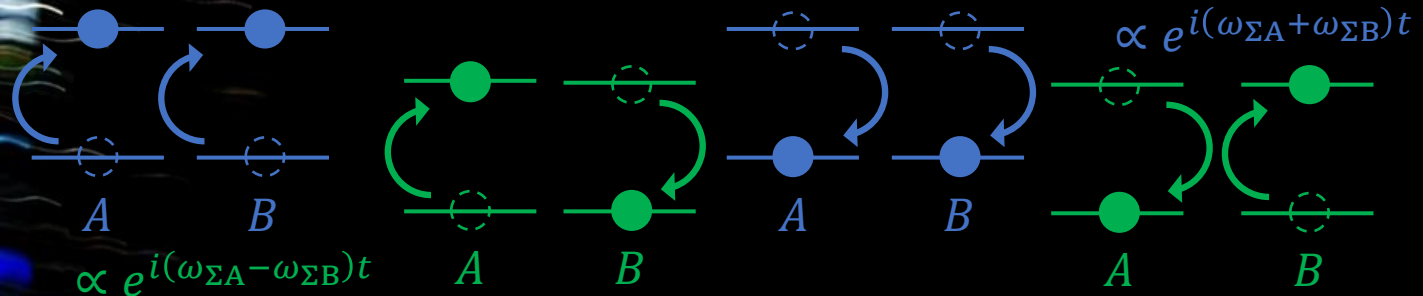
Coupled LC

$$\hat{H}_{2LC} = \frac{Q_A^2}{2C_{\Sigma A}} + \frac{L_A \Phi_A^2}{2} + \frac{Q_B^2}{2C_{\Sigma B}} + \frac{L_B \Phi_B^2}{2} - C_g \frac{Q_A}{C_A} \frac{Q_B}{C_B}$$



$$= \underbrace{\hbar\omega_{\Sigma A} \left(a_{\Sigma A}^\dagger a_{\Sigma A} + \frac{1}{2} \right) + \hbar\omega_{\Sigma B} \left(a_{\Sigma B}^\dagger a_{\Sigma B} + \frac{1}{2} \right)}_{\hat{H}_0} - \underbrace{\hbar\Omega_g i (a_{\Sigma A}^\dagger - a_{\Sigma A}) * i (a_{\Sigma B}^\dagger - a_{\Sigma B})}_{\hat{V}(t)}$$

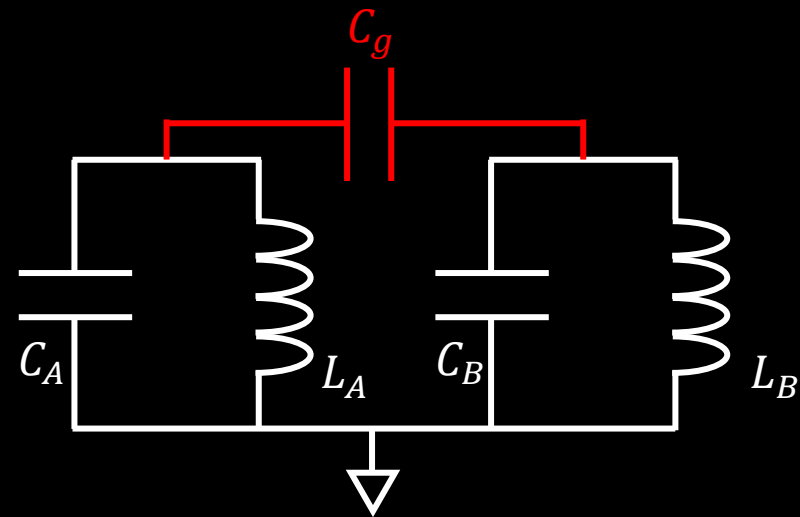
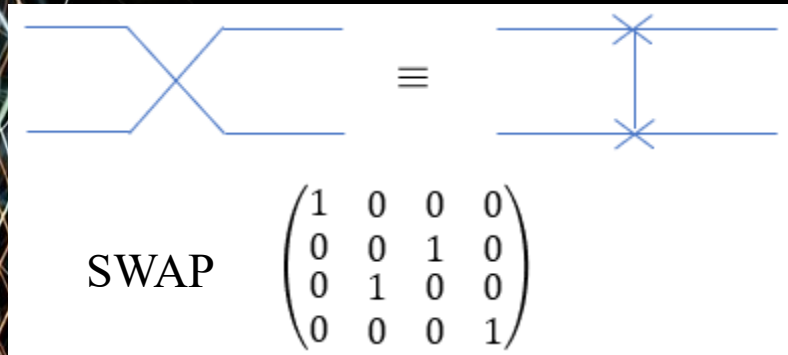
$$i\hbar \frac{\partial}{\partial t} |\Theta(t)\rangle = \hat{H}_r |\Theta(t)\rangle = \hbar\Omega_g \left[\underbrace{e^{i(\omega_{\Sigma A} + \omega_{\Sigma B})t} a_{\Sigma A}^\dagger a_{\Sigma B}^\dagger}_{\propto e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t}} + \underbrace{e^{-i(\omega_{\Sigma A} + \omega_{\Sigma B})t} a_{\Sigma A} a_{\Sigma B}}_{\propto e^{i(\omega_{\Sigma A} + \omega_{\Sigma B})t}} \right. \\ \left. - \underbrace{e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t} a_{\Sigma A}^\dagger a_{\Sigma B}}_{\propto e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t}} - \underbrace{e^{-i(\omega_{\Sigma A} - \omega_{\Sigma B})t} a_{\Sigma A} a_{\Sigma B}^\dagger}_{\propto e^{i(\omega_{\Sigma A} + \omega_{\Sigma B})t}} \right] |\Theta(t)\rangle$$



Rotating Wave Approximation (RWA)

1. $\omega \sim \text{GHz}$, detuning $\Delta \equiv \omega_{\Sigma A} - \omega_{\Sigma B} \sim 50 \text{ MHz}$, $2\omega \gg \Delta$ too fast
2. Transition $|i\rangle \rightarrow |f\rangle \propto -\frac{i}{\hbar} \int_0^t \langle f | \hat{H}_r | i \rangle dt \sim \frac{\dots}{\omega_{\Sigma A} + \omega_{\Sigma B}}$ or $\frac{\dots}{\omega_{\Sigma A} - \omega_{\Sigma B}}$

SWAP (iSWAP)



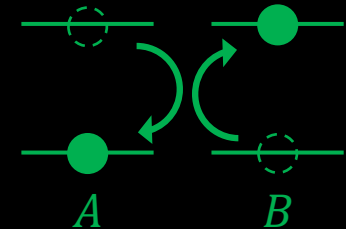
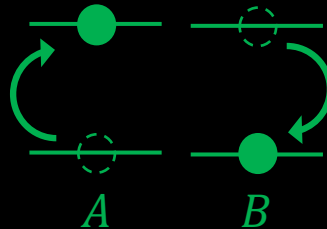
$$|0_A 0_B\rangle \rightarrow |0_A 0_B\rangle$$

$$|1_A 1_B\rangle \rightarrow |1_A 1_B\rangle$$

$$|0_A 1_B\rangle \rightarrow i|1_A 0_B\rangle$$

$$|1_A 0_B\rangle \rightarrow i|0_A 1_B\rangle$$

$$\propto e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t}$$

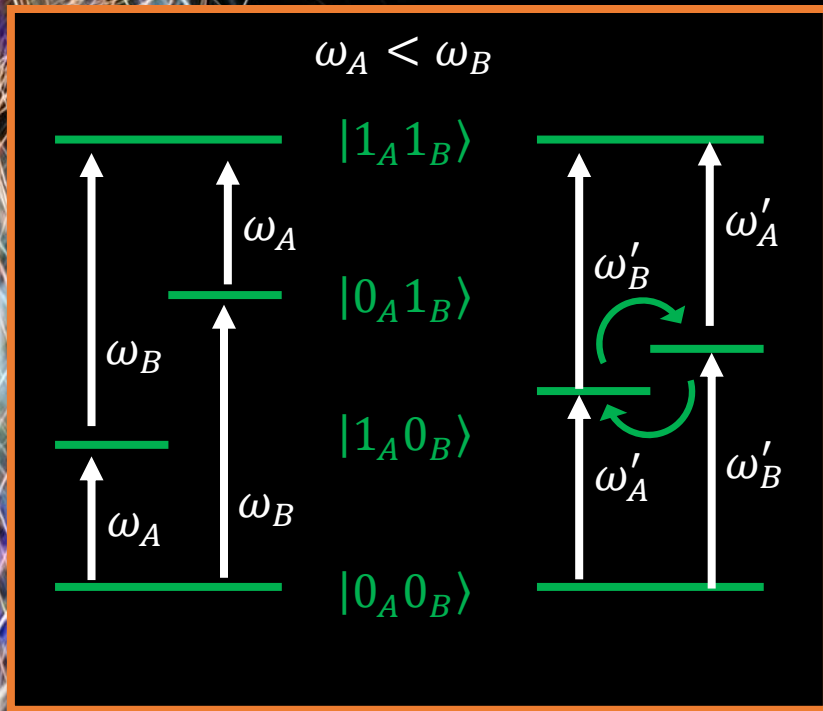


Rotating Wave Approximation (RWA)

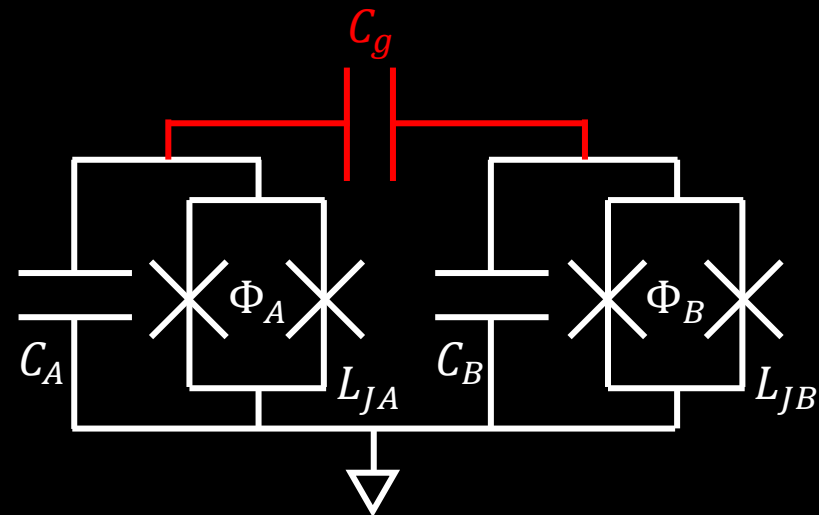
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SWAP (iSWAP)



$$\propto e^{i(\omega_{\Sigma A} - \omega_{\Sigma B})t}$$

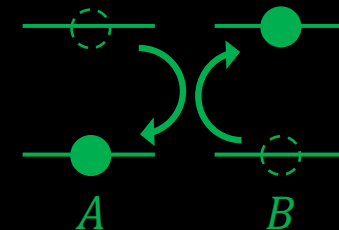
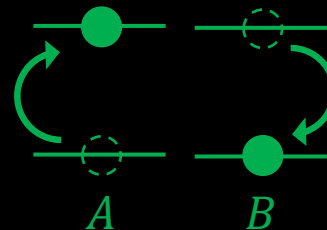


$$|0_A 0_B\rangle \rightarrow |0_A 0_B\rangle$$

$$|1_A 1_B\rangle \rightarrow |1_A 1_B\rangle$$

$$|0_A 1_B\rangle \rightarrow i|1_A 0_B\rangle$$

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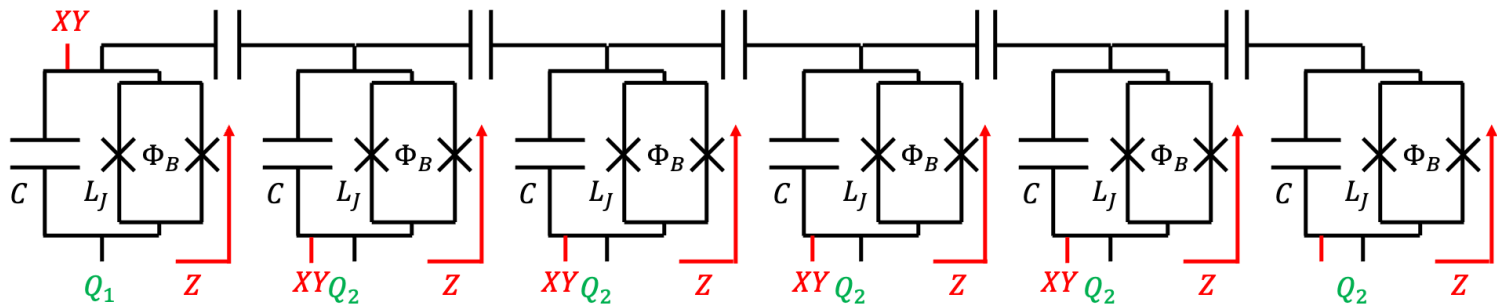
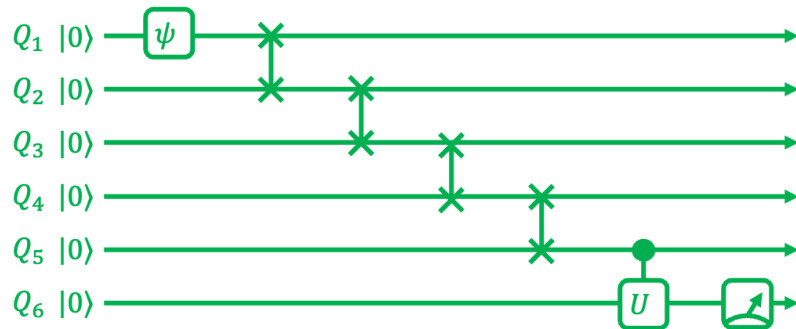


Rotating Wave Approximation (RWA)

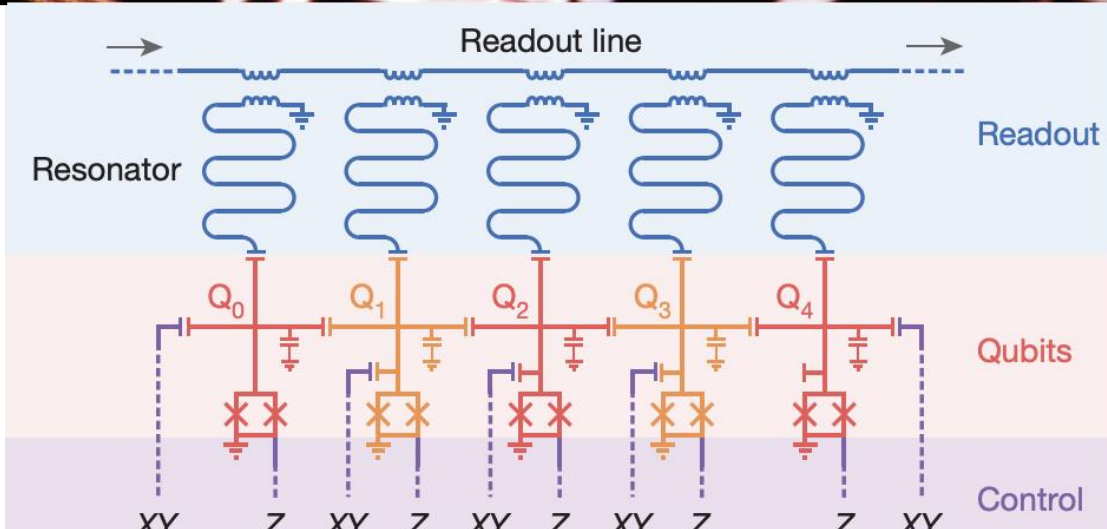
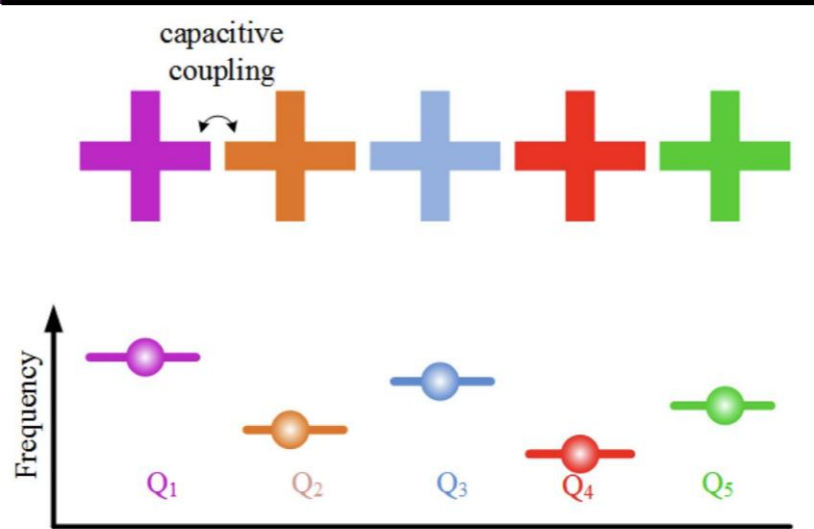
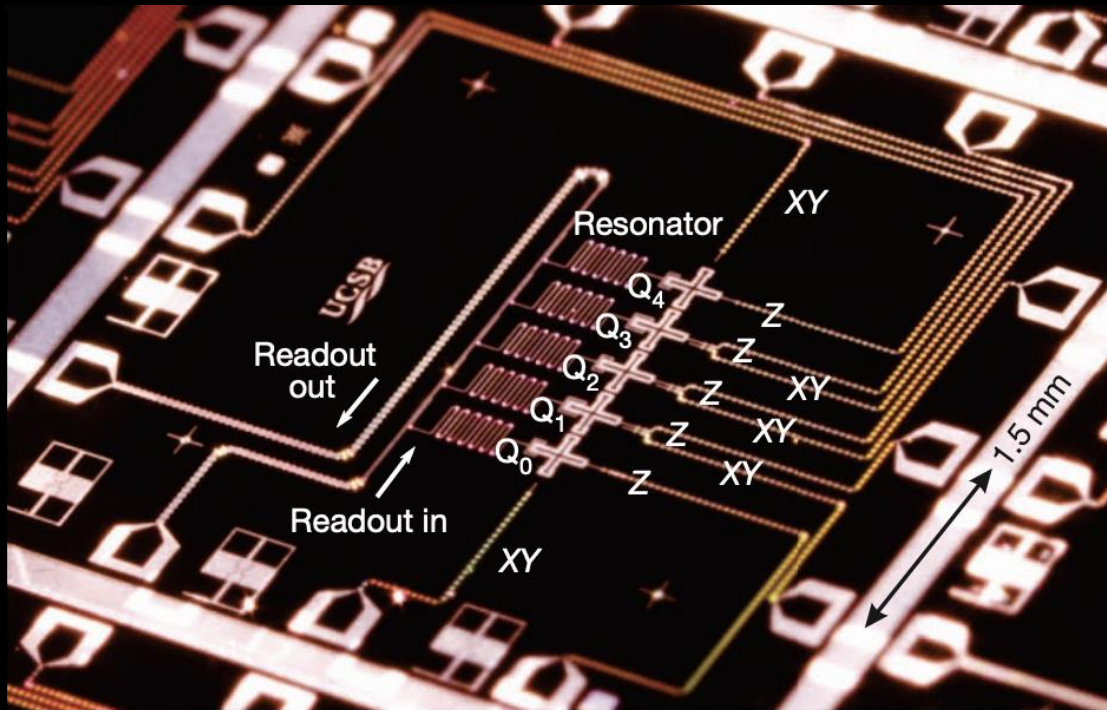
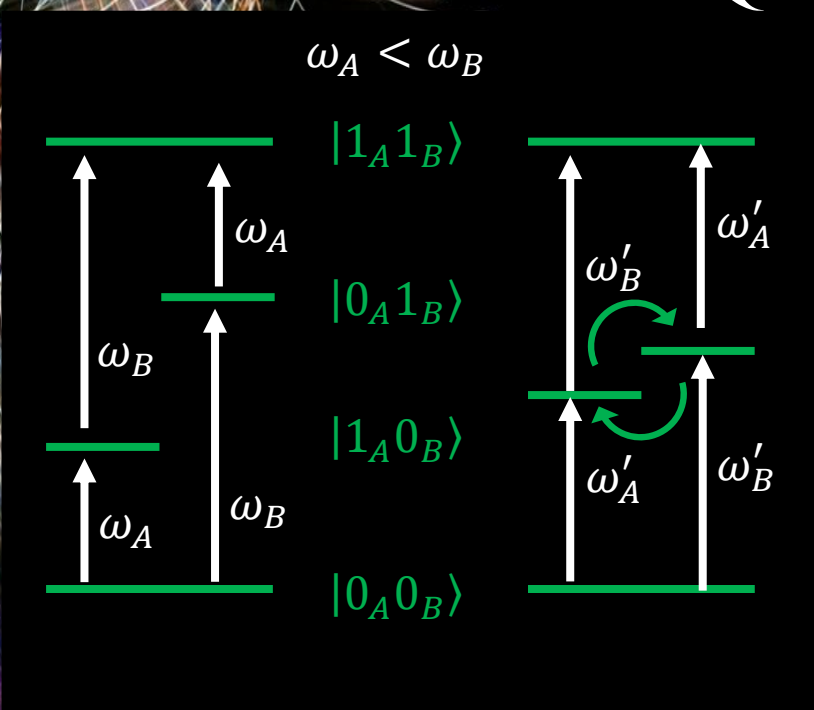
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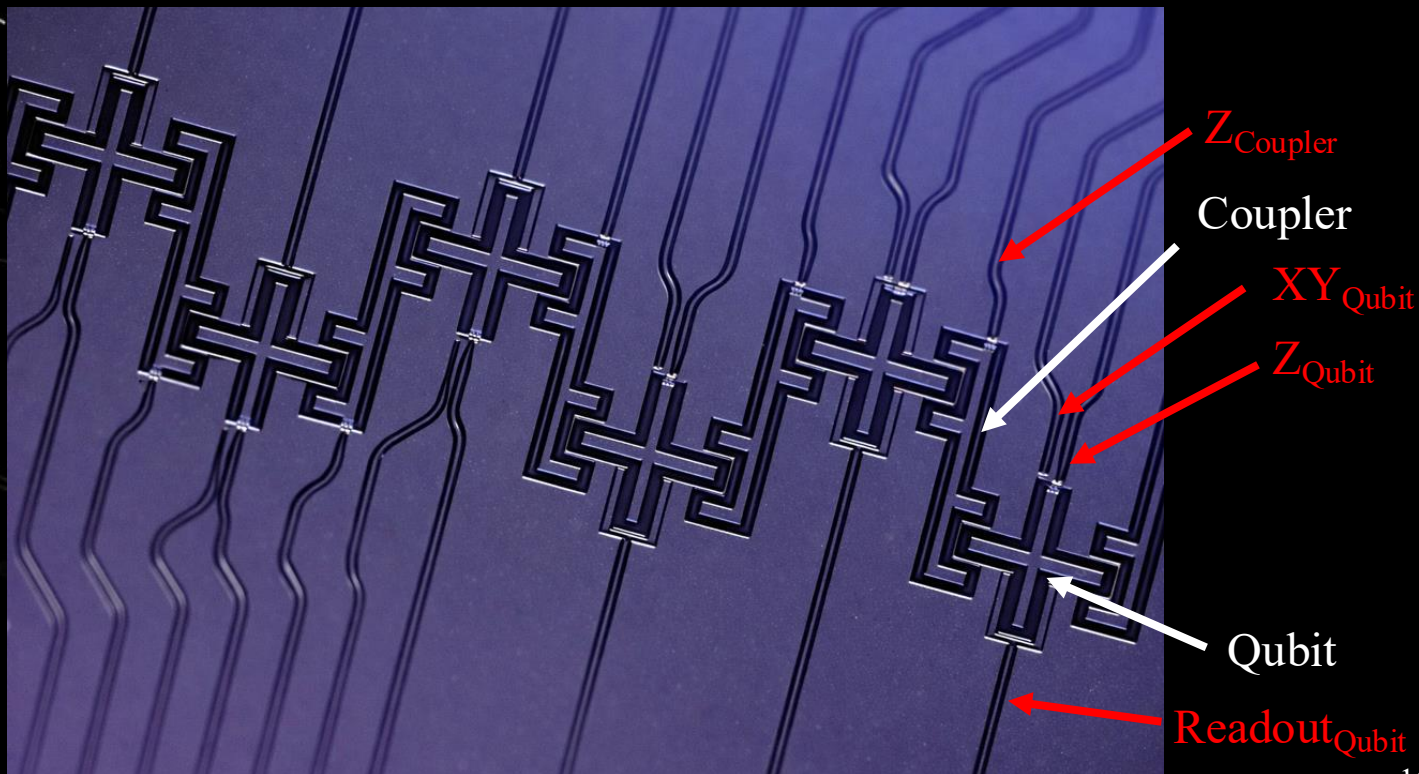
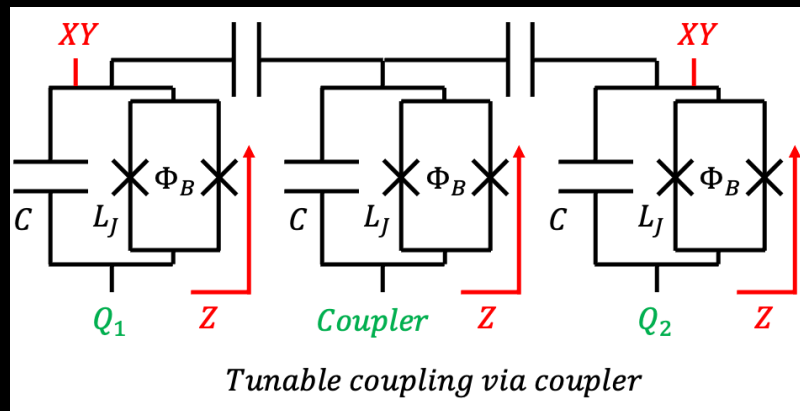
SWAP (iSWAP)

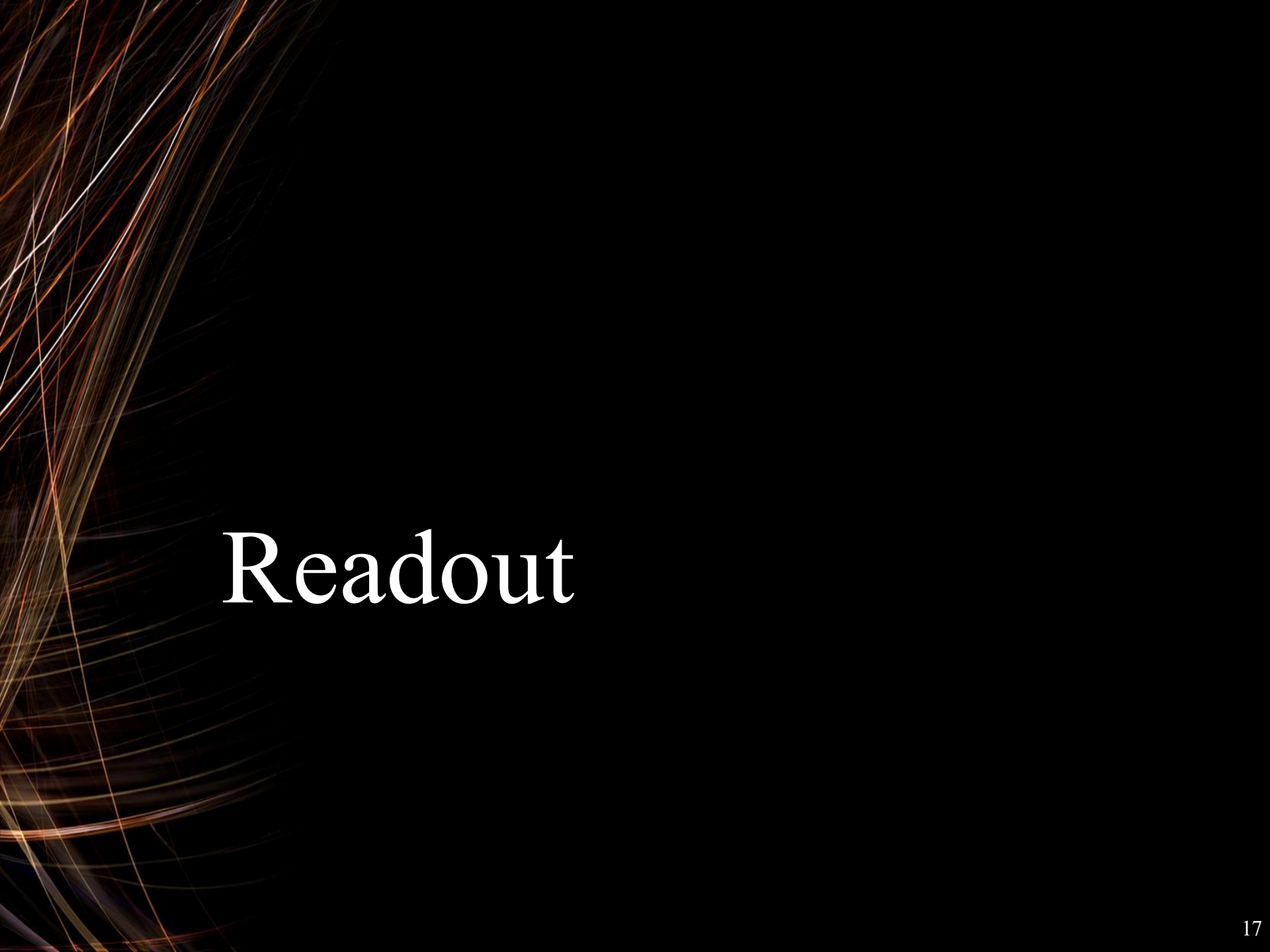


Multi Qubit Circuit



Tunable Coupler



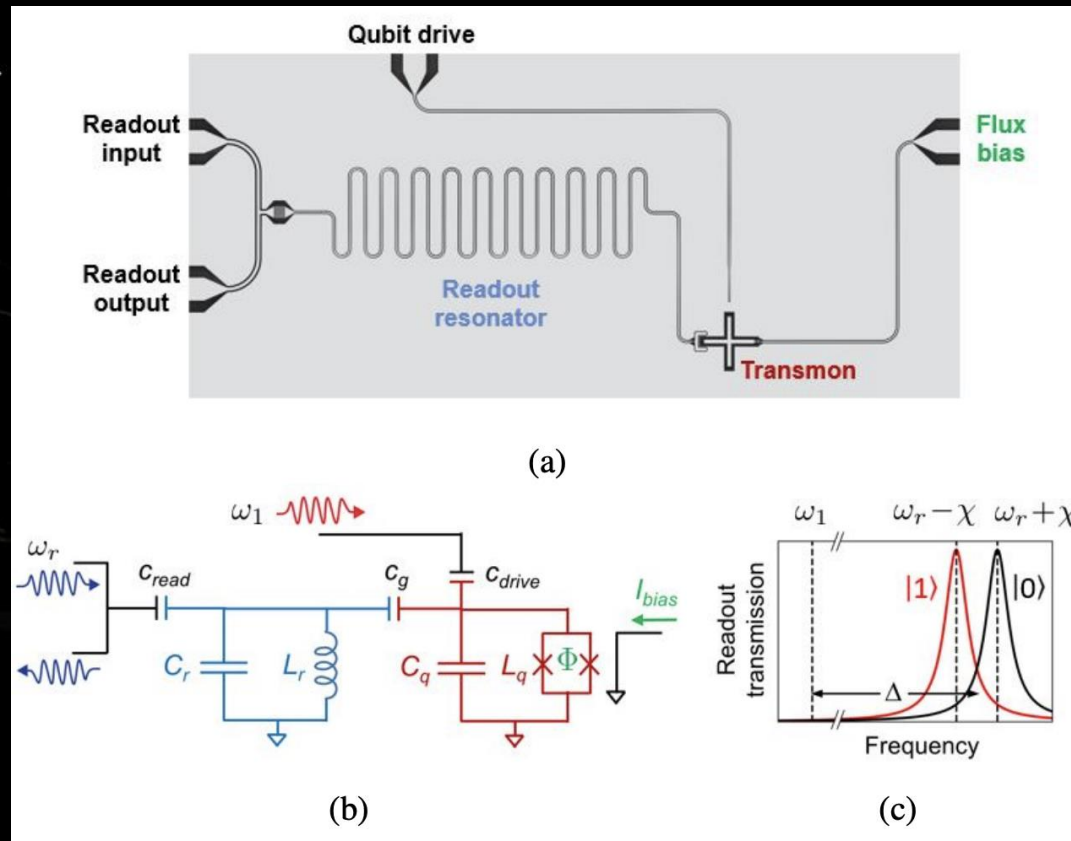
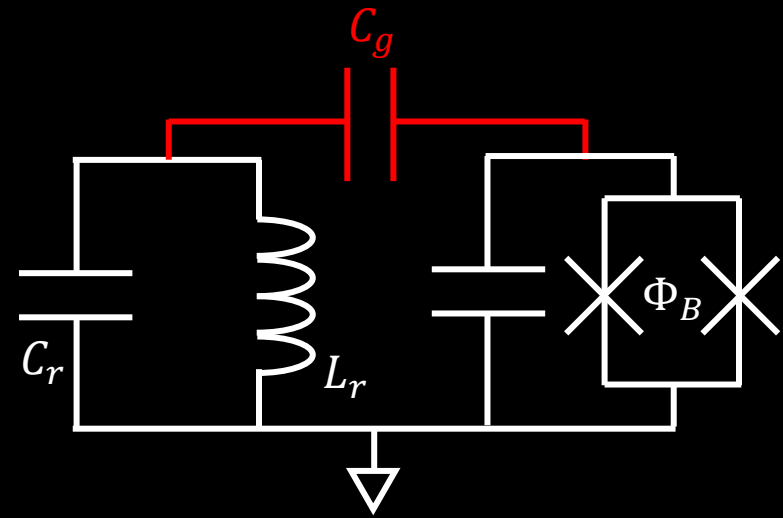


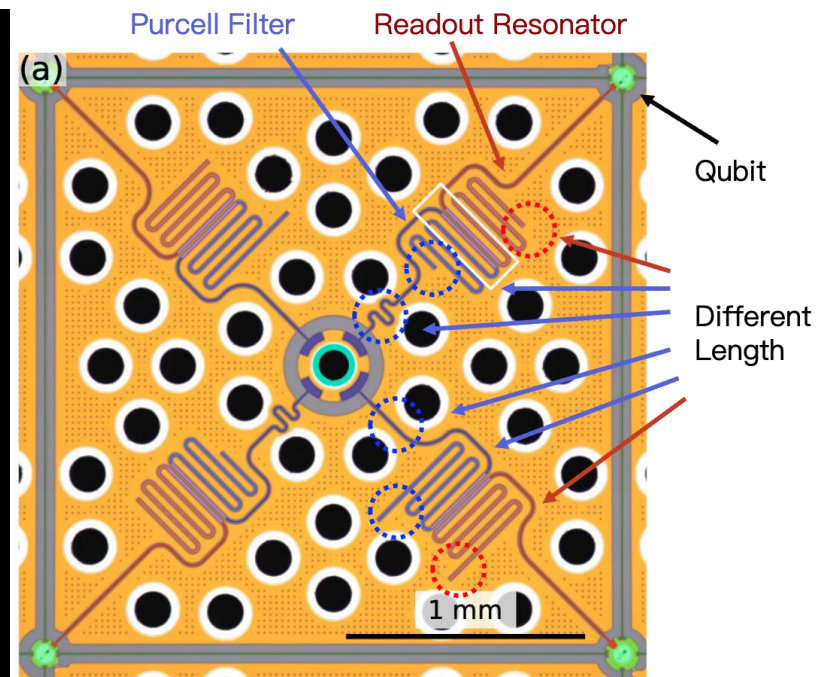
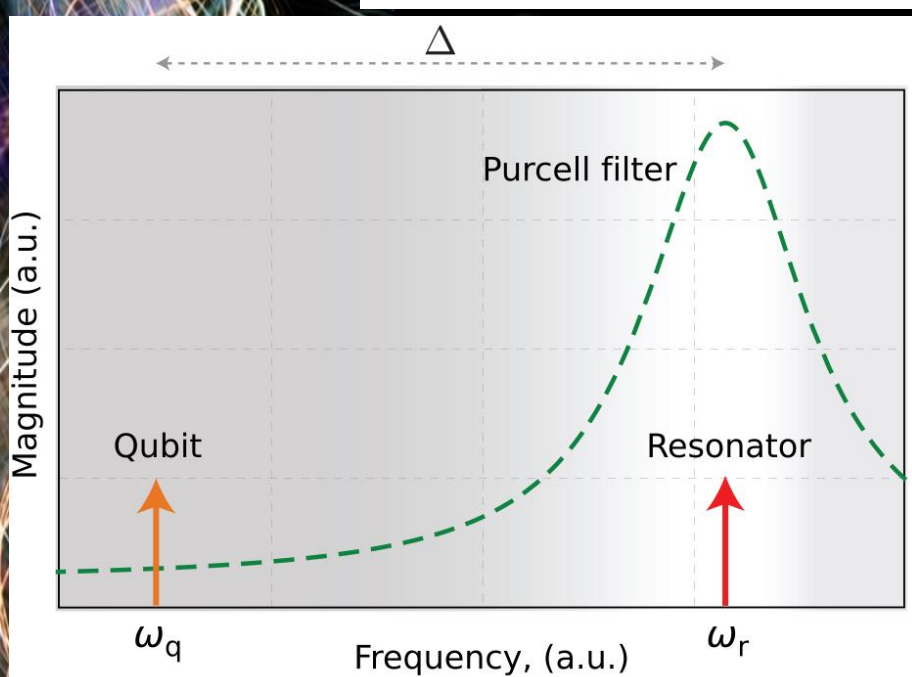
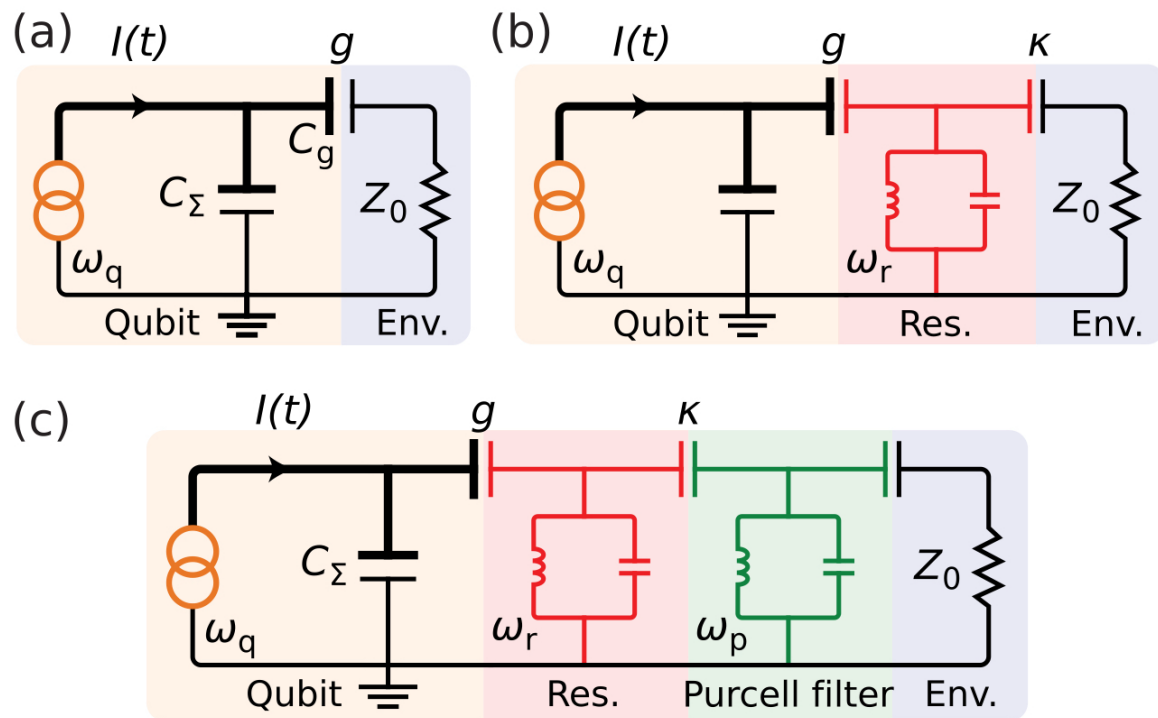
Readout

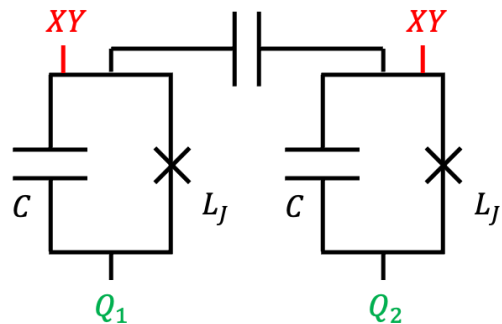
AC Stark Effect

$$H \cong \hbar\omega_r a_r^\dagger a_r - 2\chi a_r^\dagger a_r * \sigma_q^\dagger \sigma_q + \hbar\omega_q \sigma_q^\dagger \sigma_q$$

$$= \hbar(\omega_r - 2\chi\sigma_q^\dagger \sigma_q) a_r^\dagger a_r + \hbar\omega_q \sigma_q^\dagger \sigma_q$$

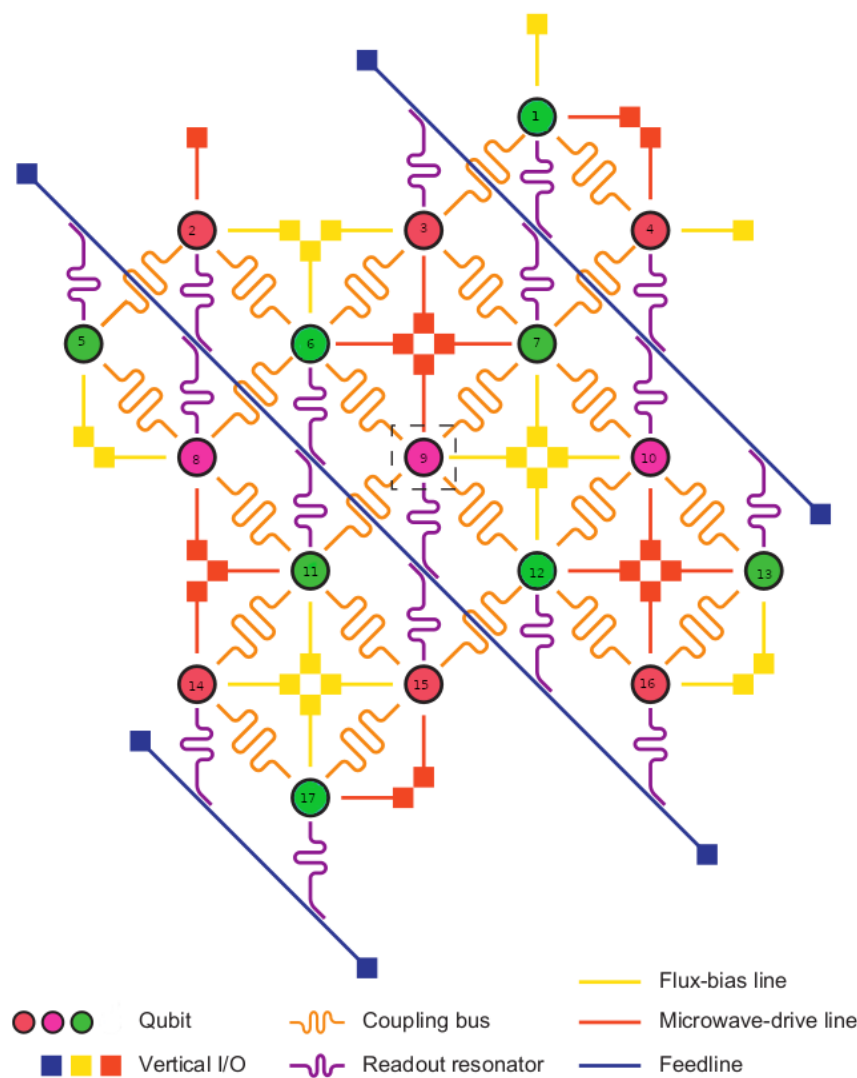
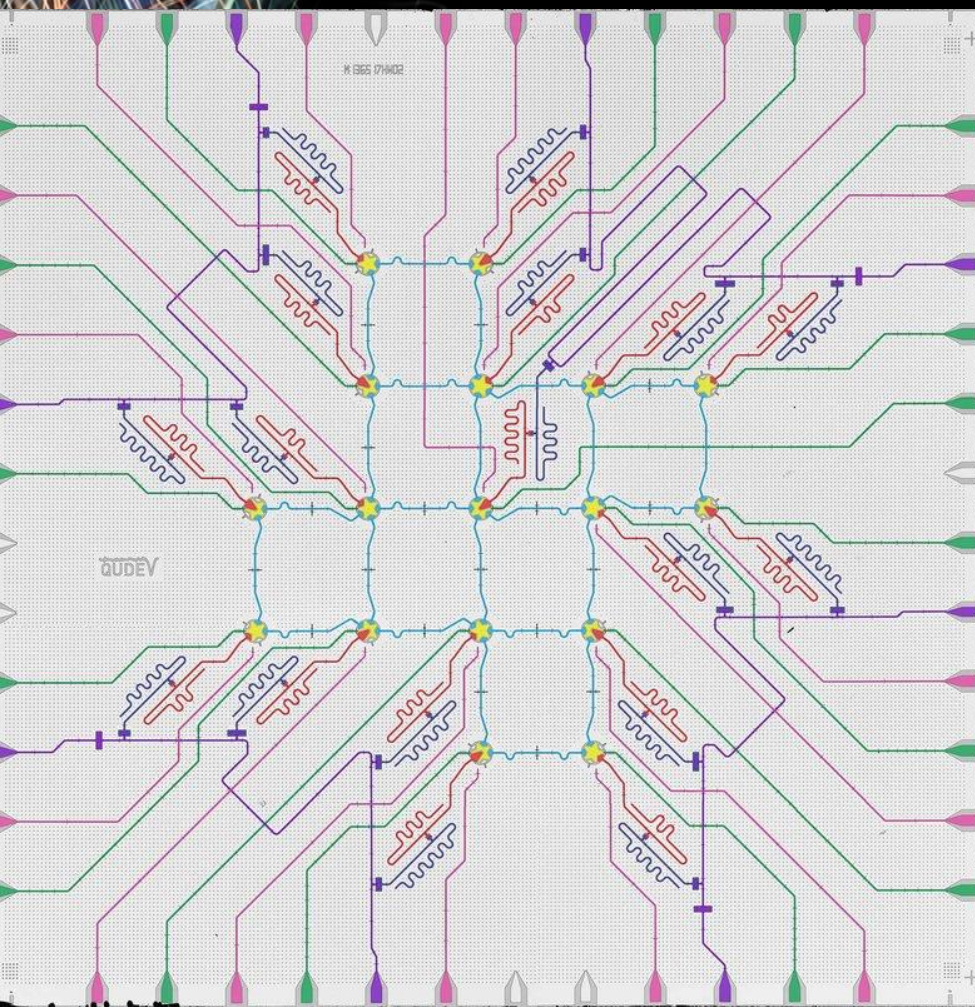






Direct fixed coupling

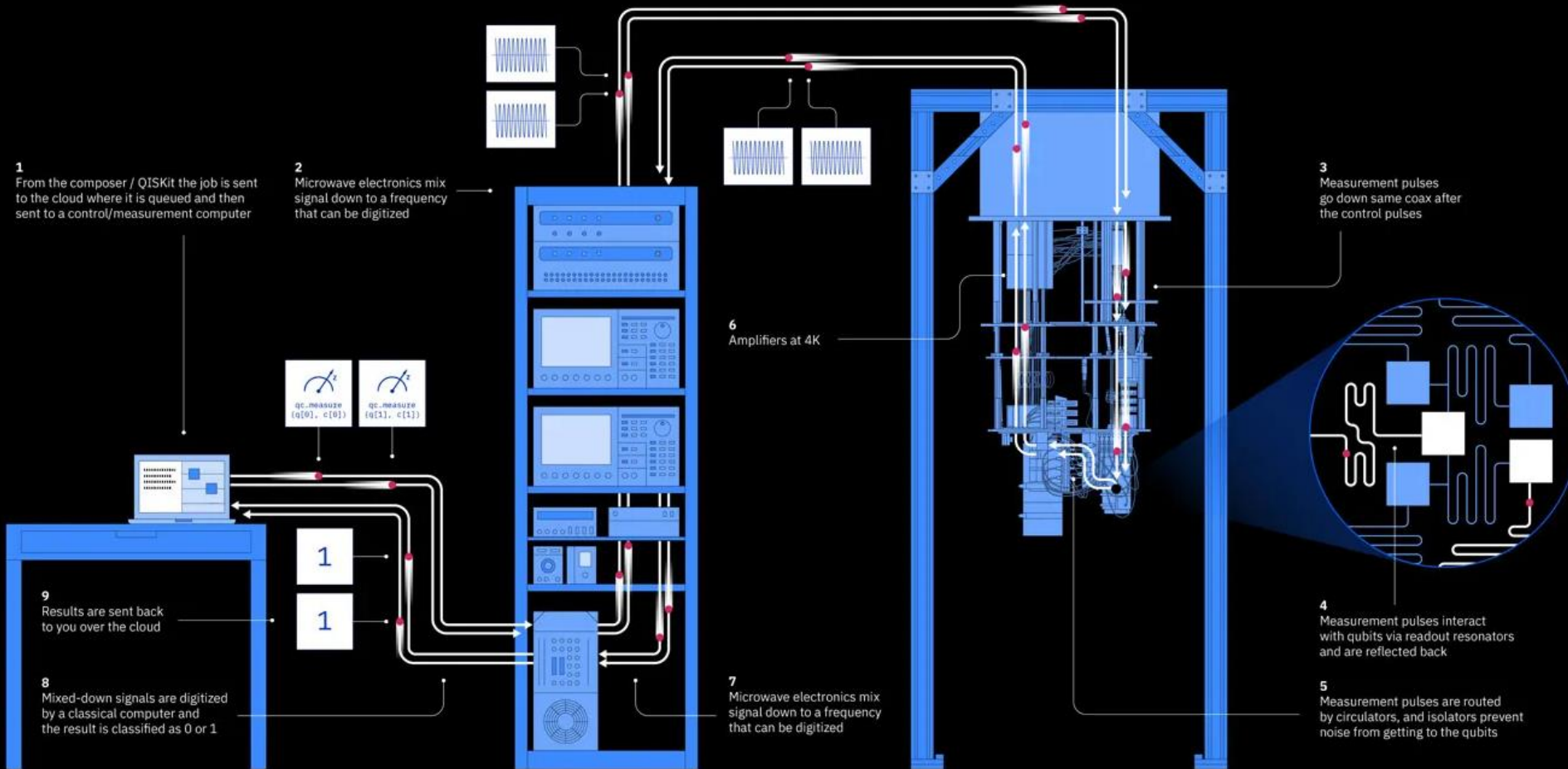
<https://www.nextbigfuture.com/2021/12/quantum-computer-error-correction-advance.html>
https://www.researchgate.net/figure/QuTech-Surface-17-S-17-Quantum-Processor-Figure-edited-from-6_fig1_364126234



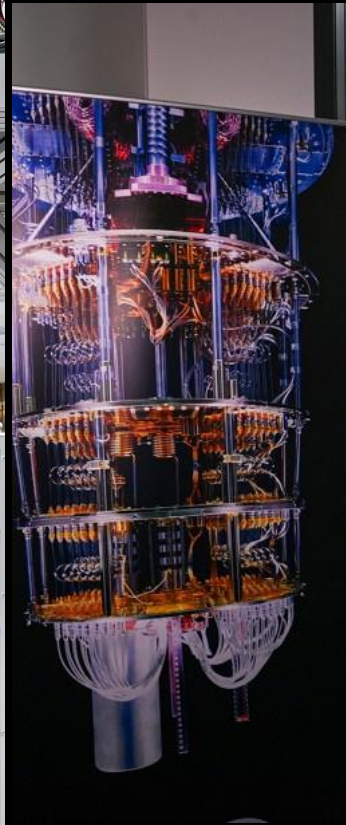


Hardware

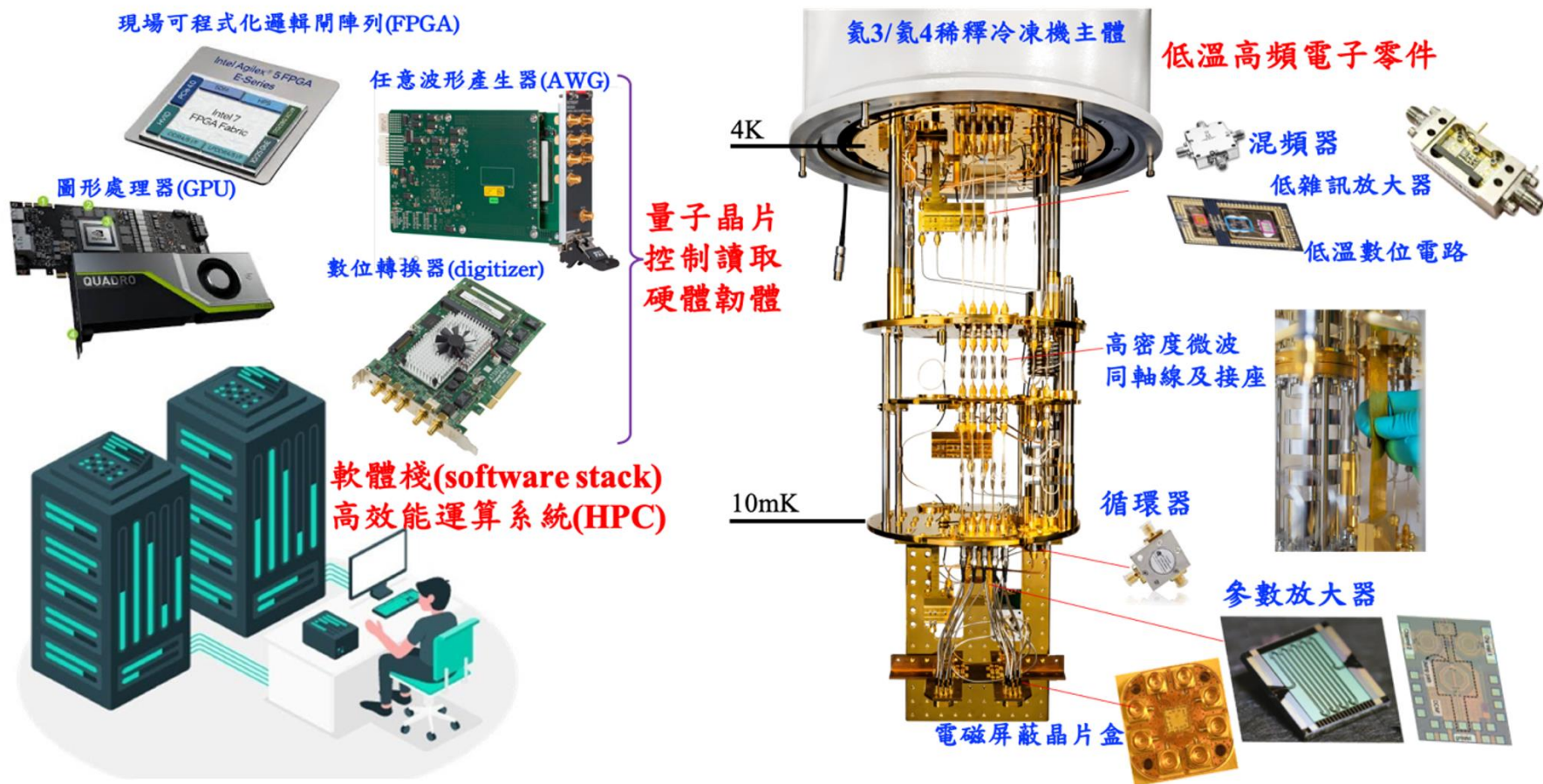
IBM's Prototype

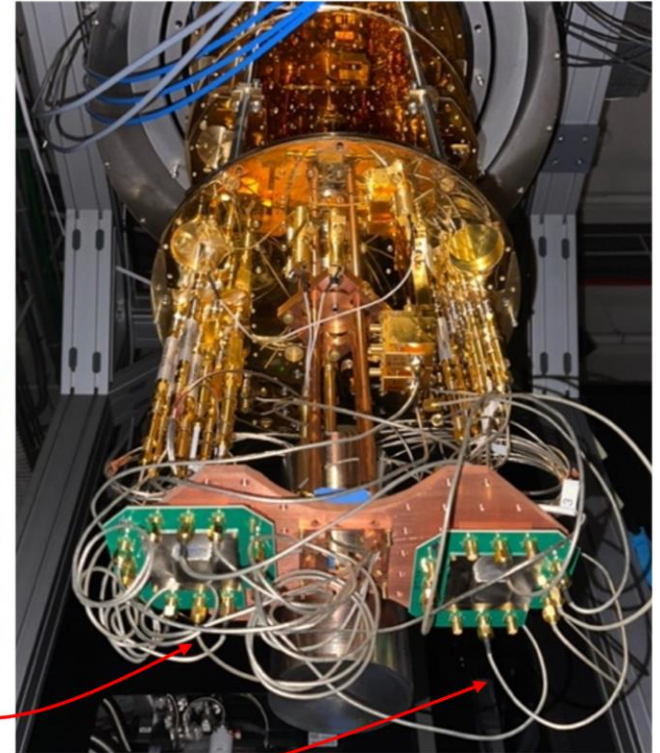
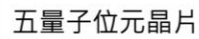


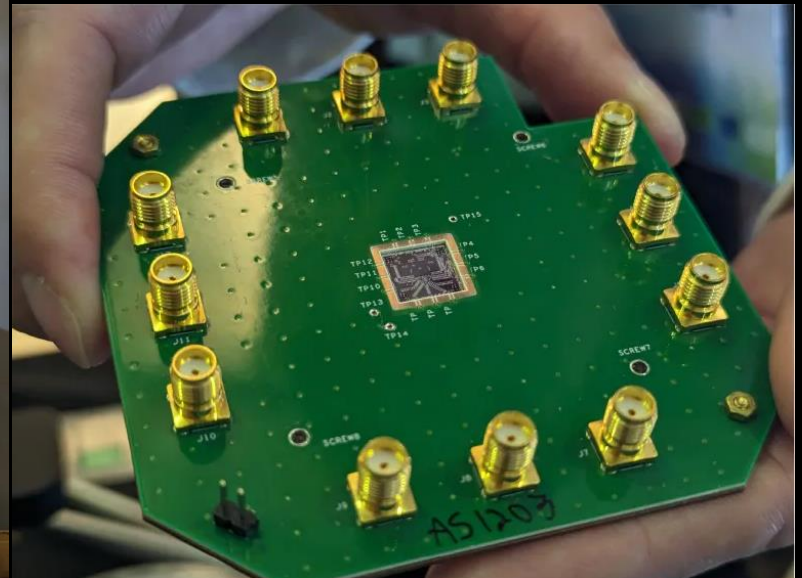
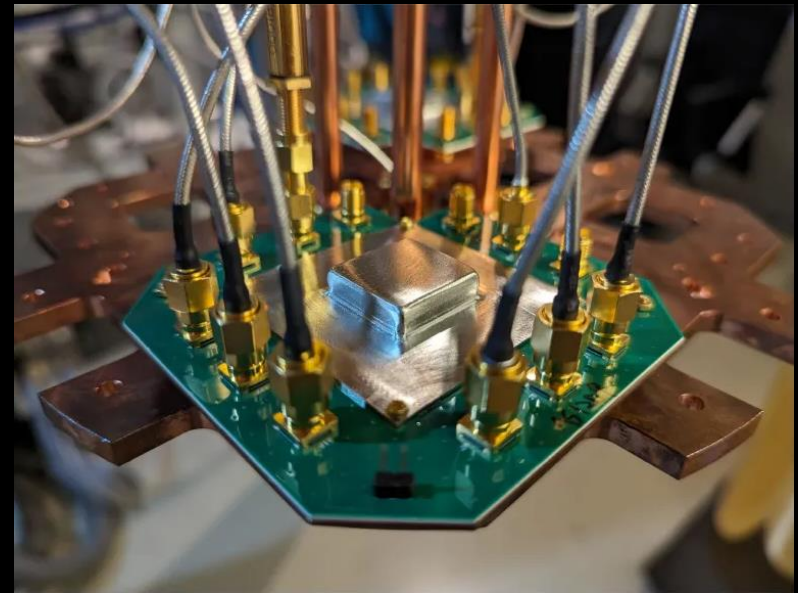
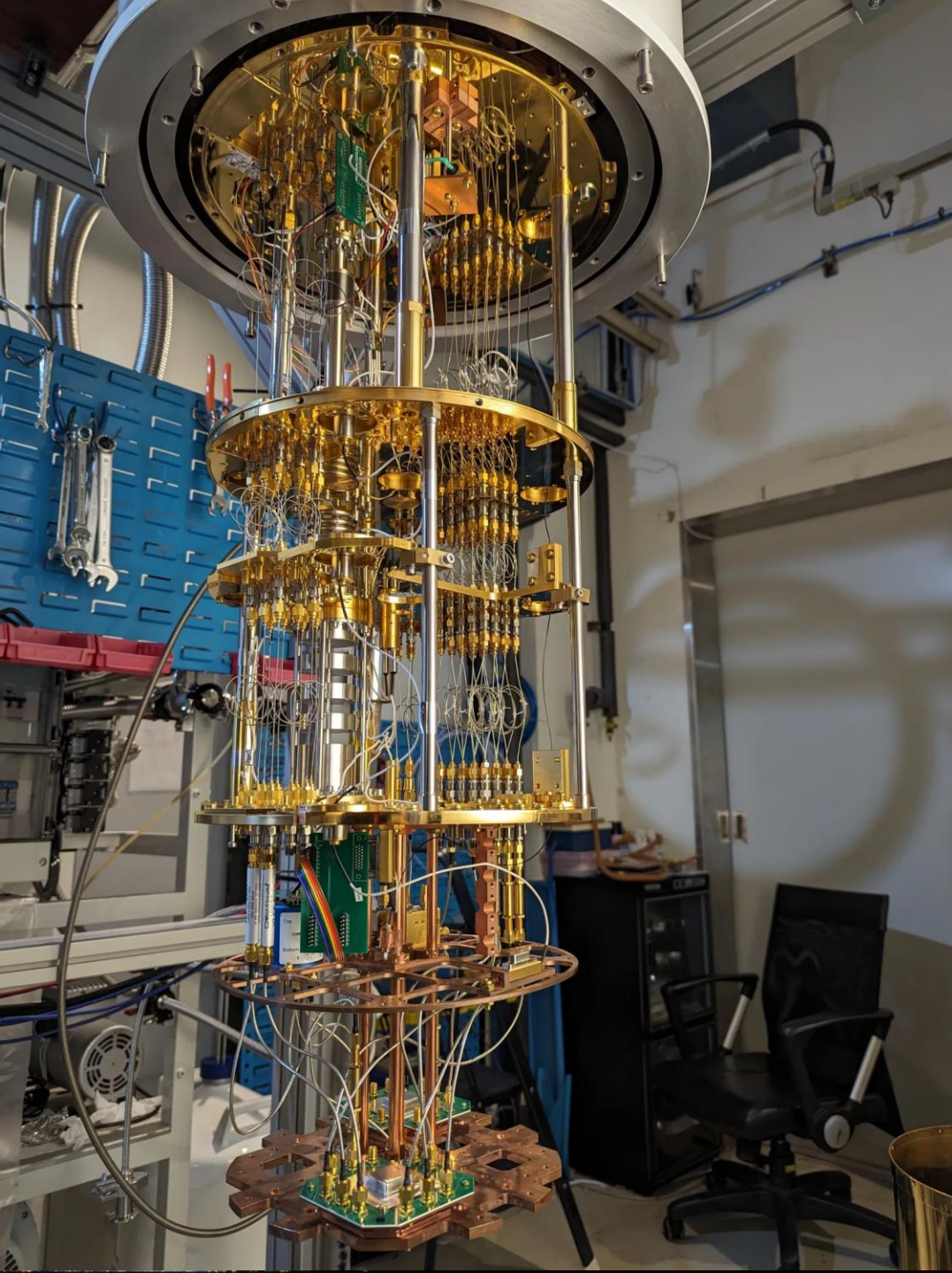
Prototype

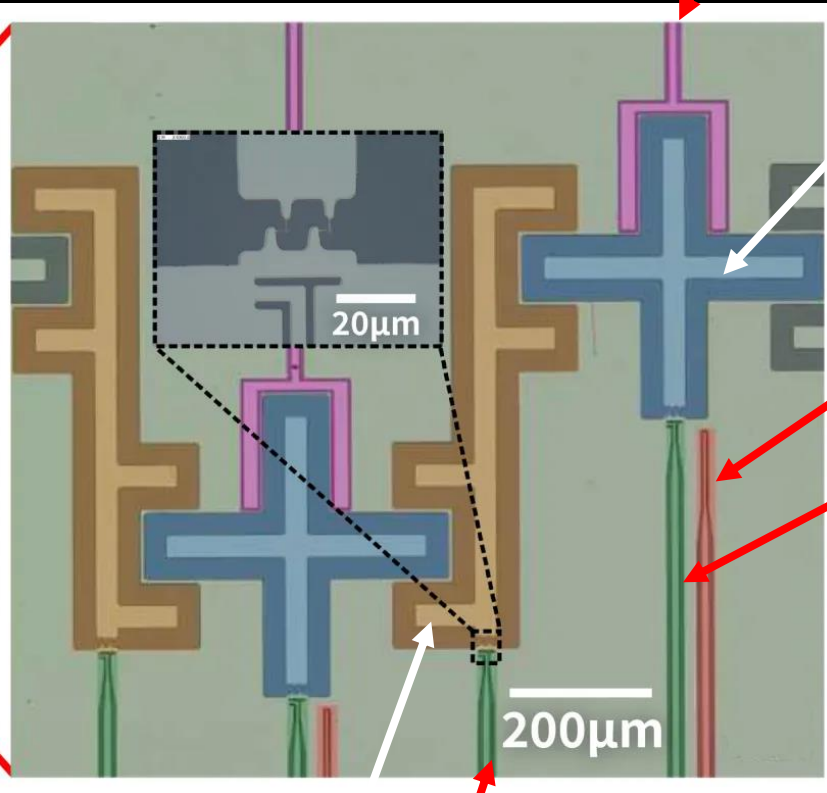
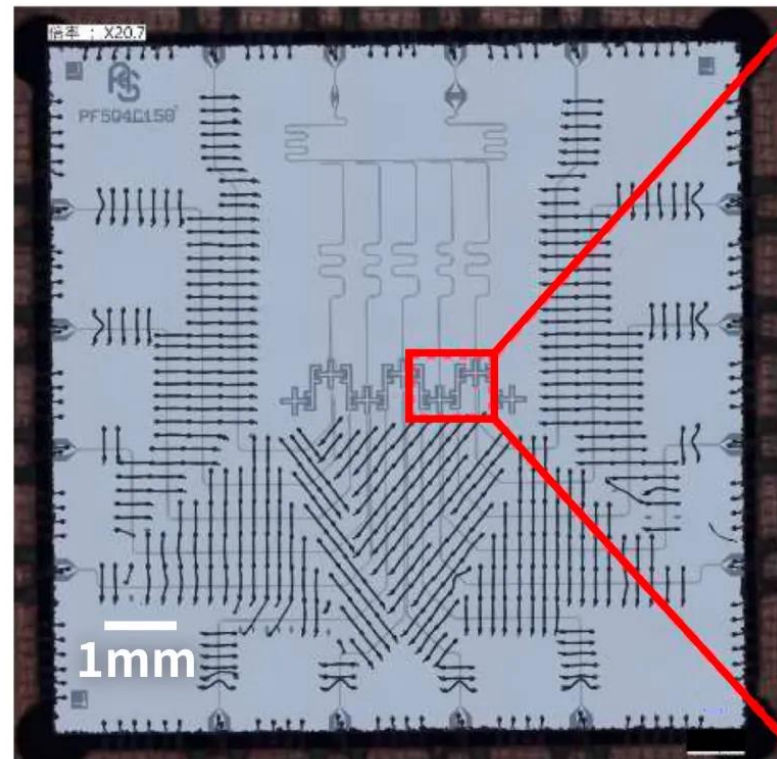
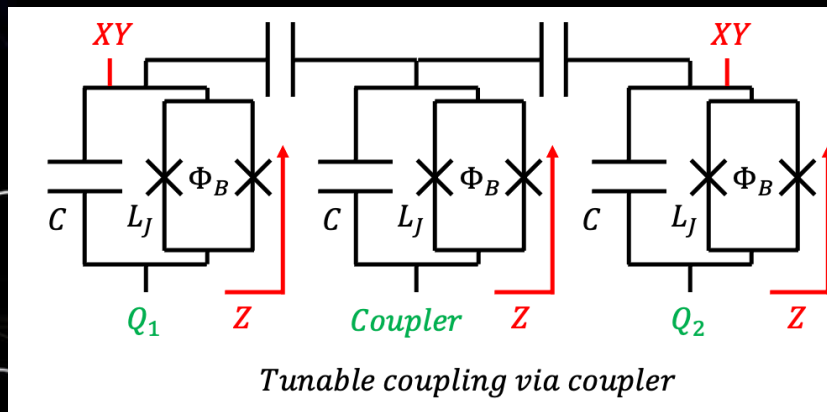


Sinica 5 Qubit









Readout

Qubit

XY_{Qubit}

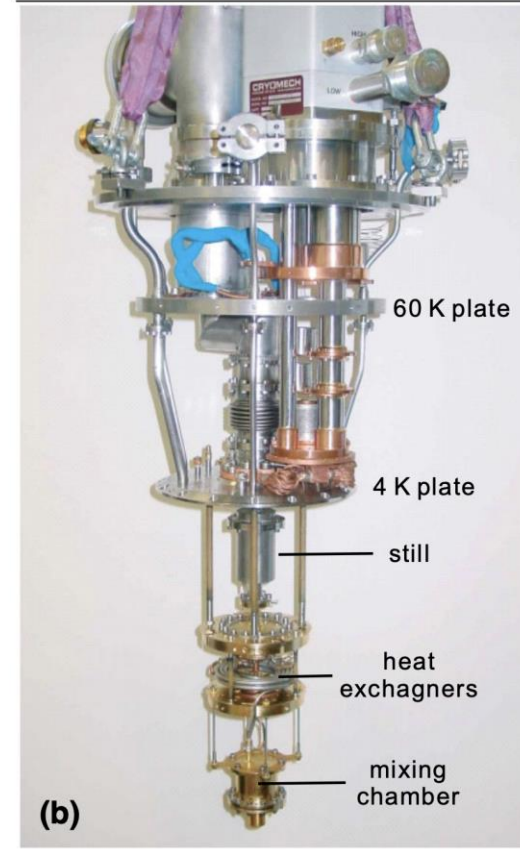
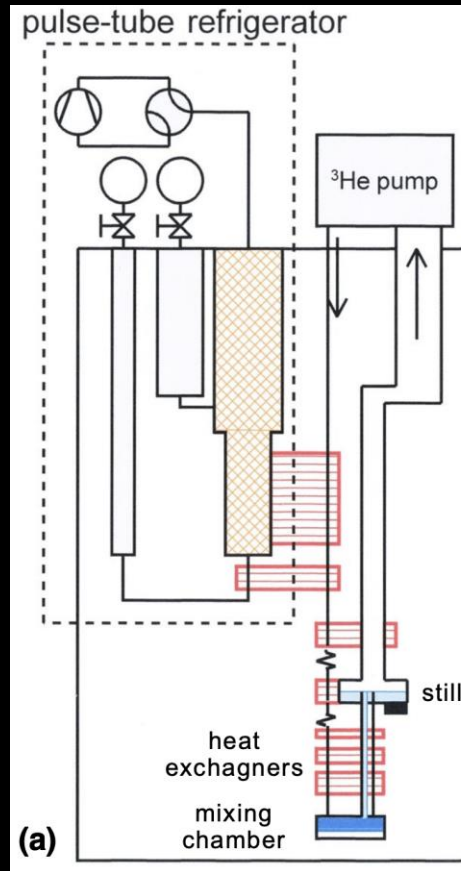
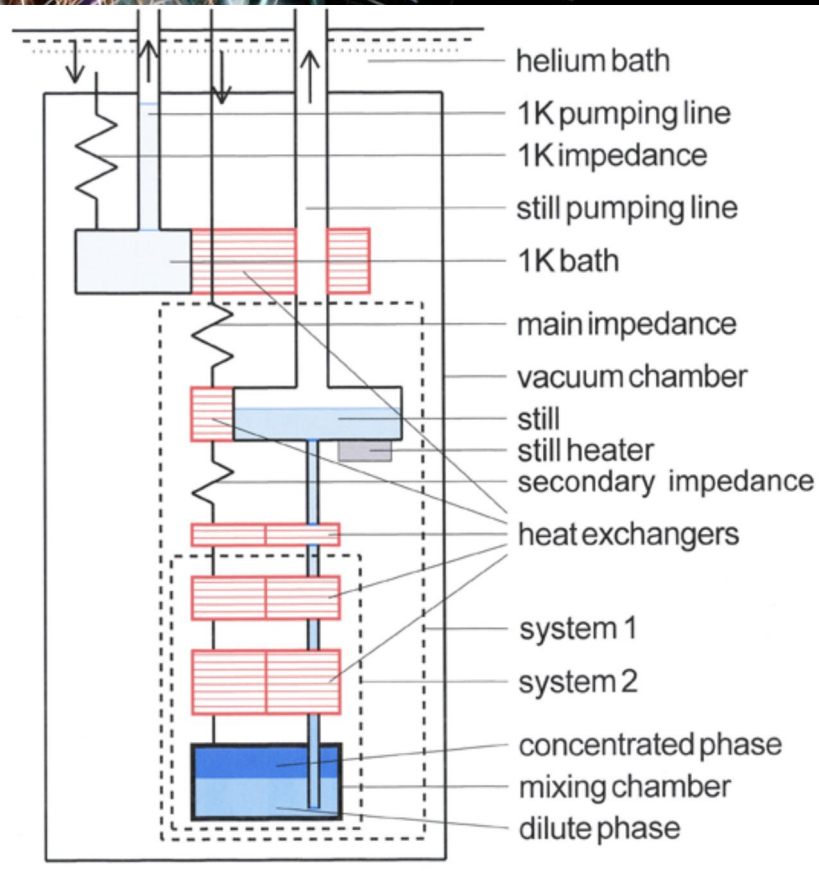
Z_{Qubit}

Coupler

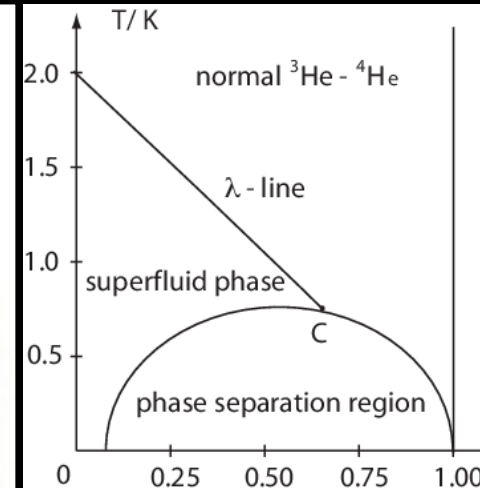
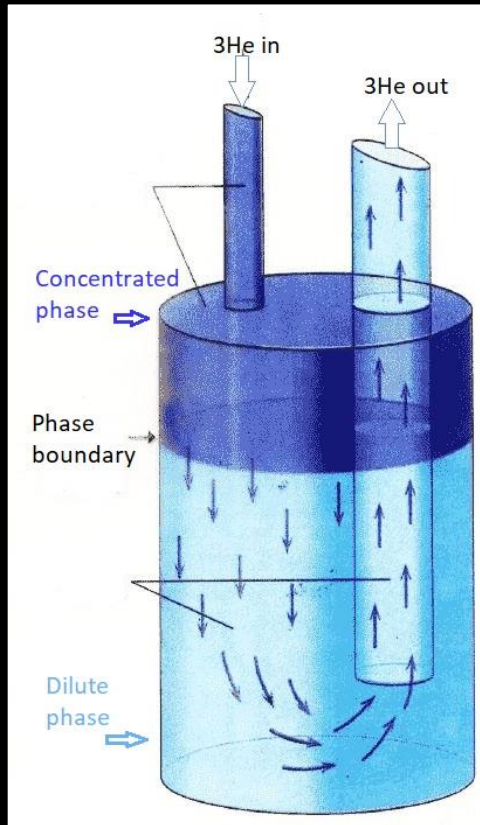
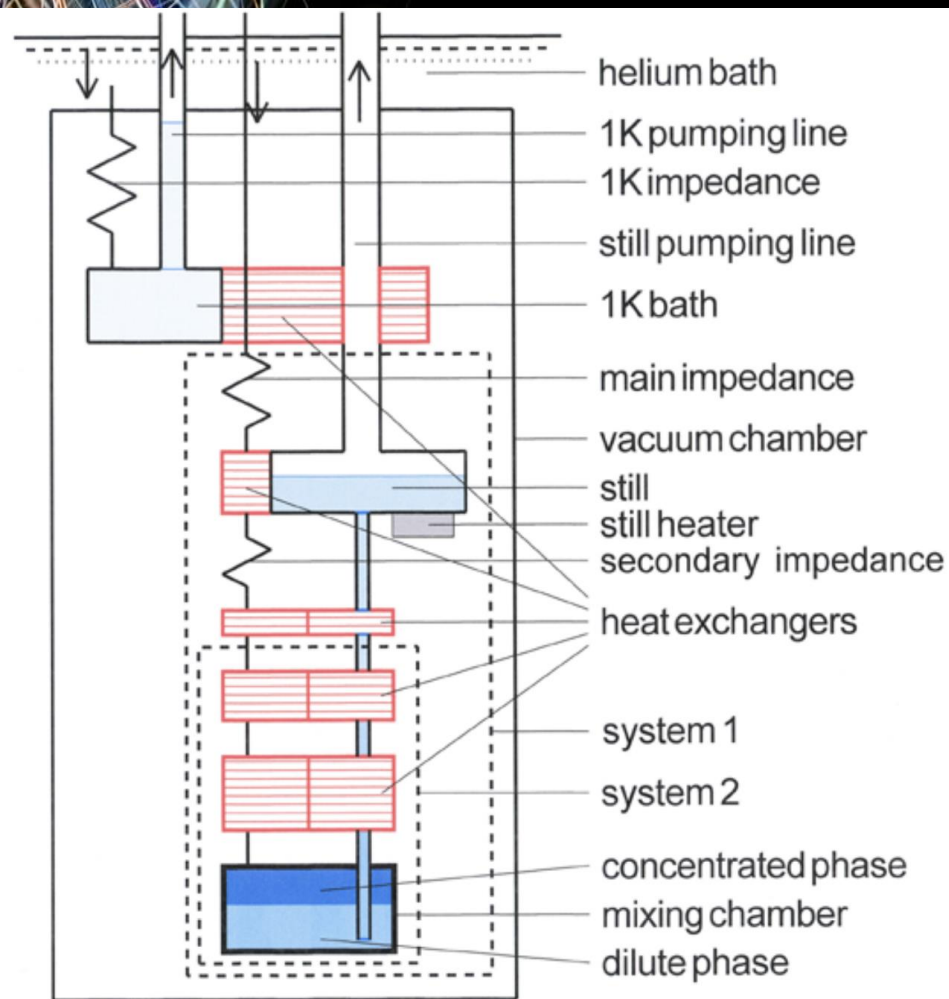
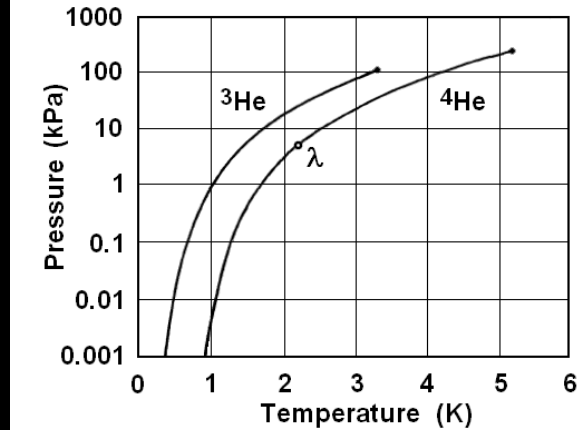
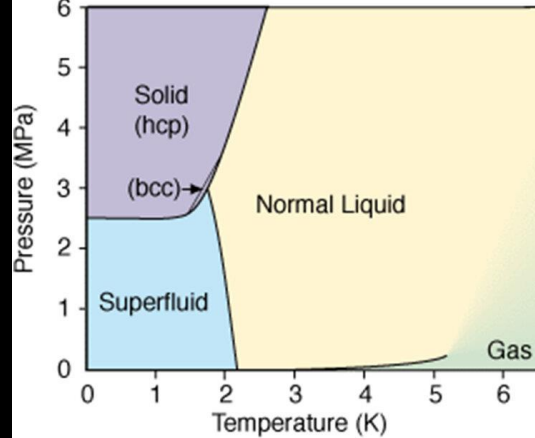
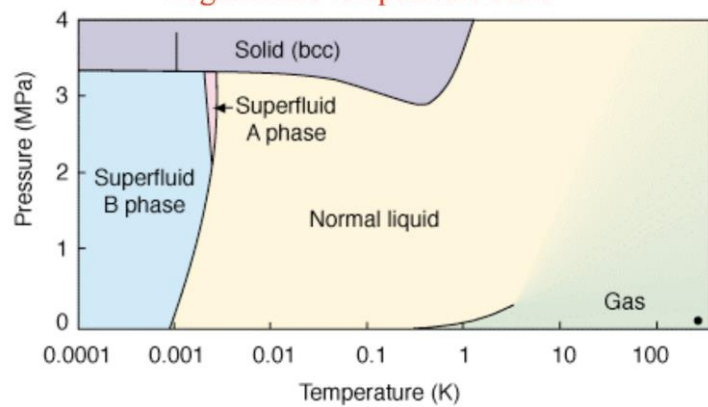
$Z_{Coupler}$

Wet and Dry Dilution Fridge

濕式/乾式稀釋製冷機

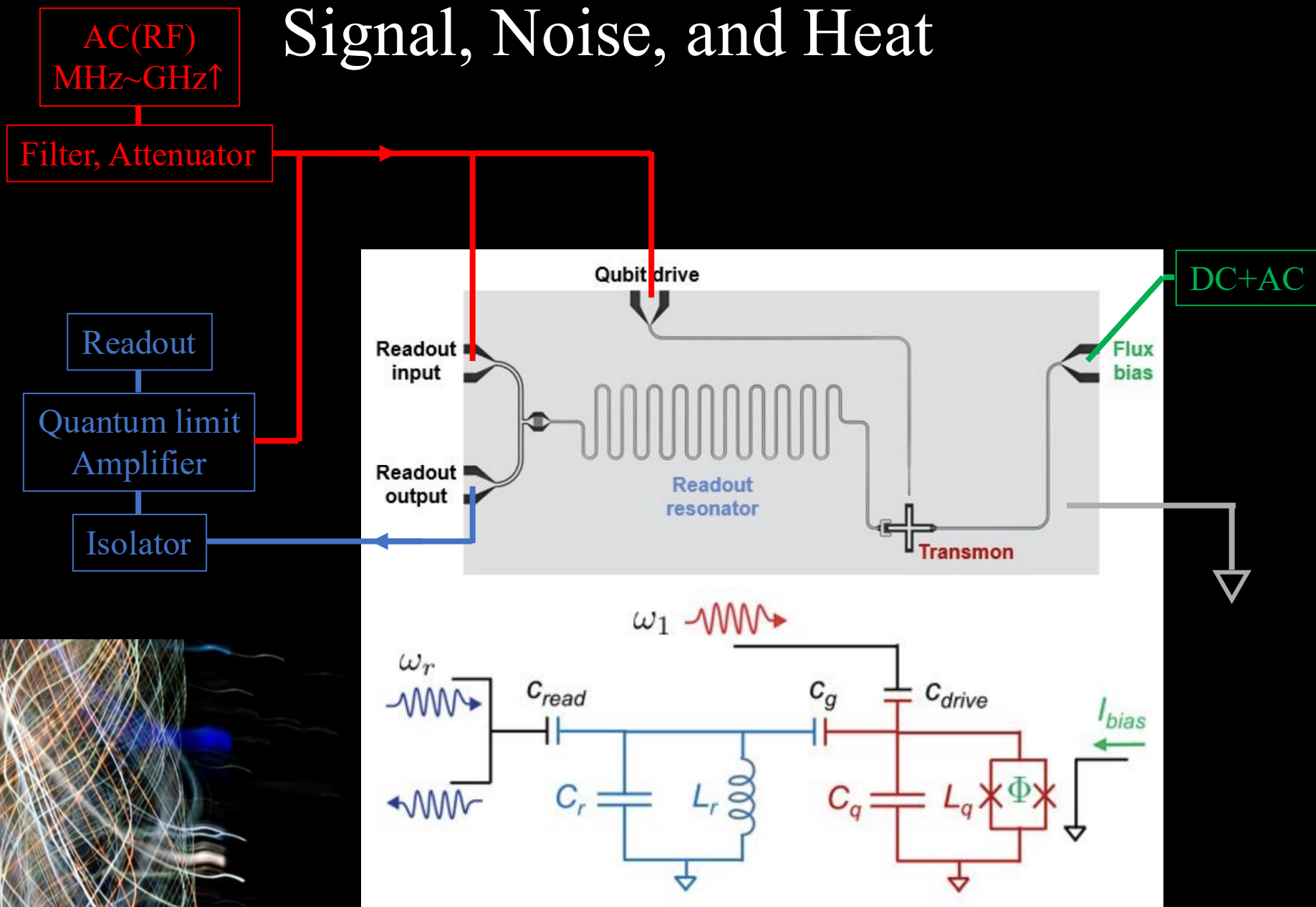


Logarithmic temperature scale

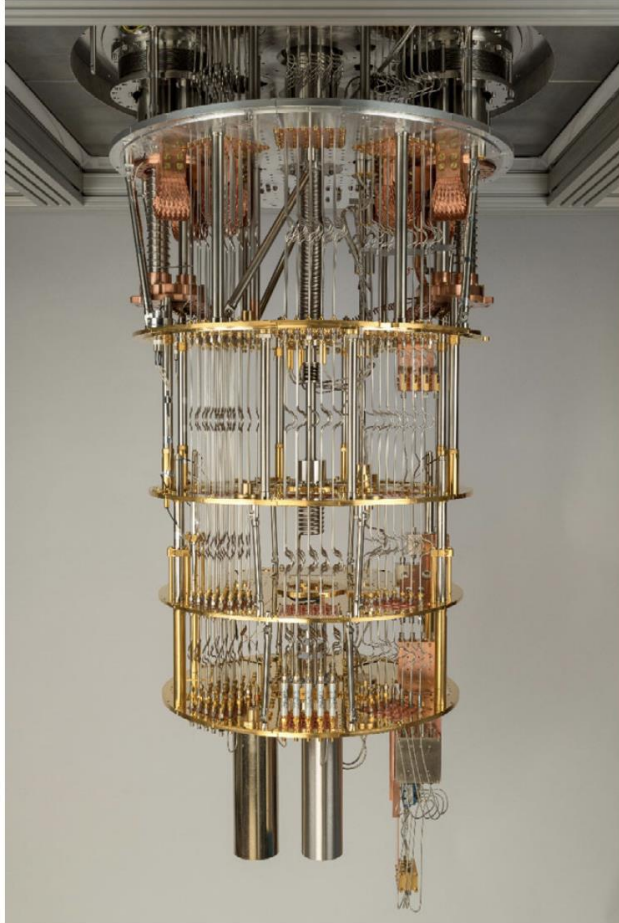


https://www.researchgate.net/figure/liquid-mixture-phase-diagram-of-3-He-4-He_fig1_1733936
https://commons.wikimedia.org/wiki/File:3He_4He_saturated_vapor_pressure.gif
[doi:10.1007/s10909-021-02606-7](https://doi.org/10.1007/s10909-021-02606-7)
https://emp.kip.uni-heidelberg.de/crytools/doku.php?id=wikidesign_construction
 ^3He Ref: <https://fy.chalmers.se/~delsing/LowTemp/LectureHc3/Helium-3.pdf>
 ^4He Ref: DOI:10.1088/1361-6633/ab3df5

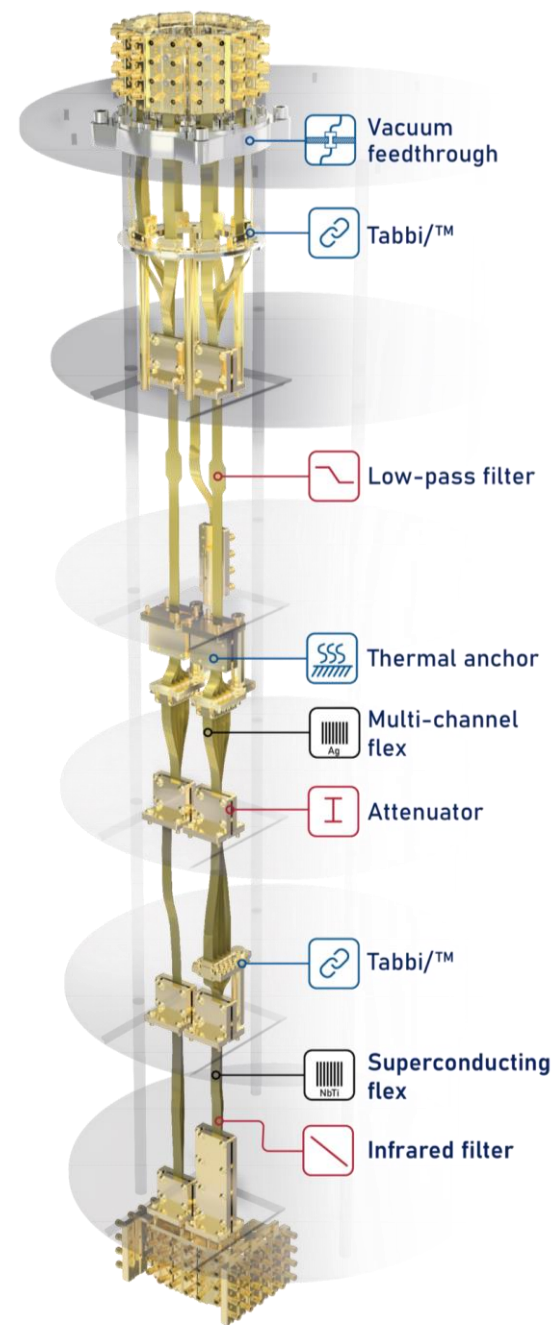
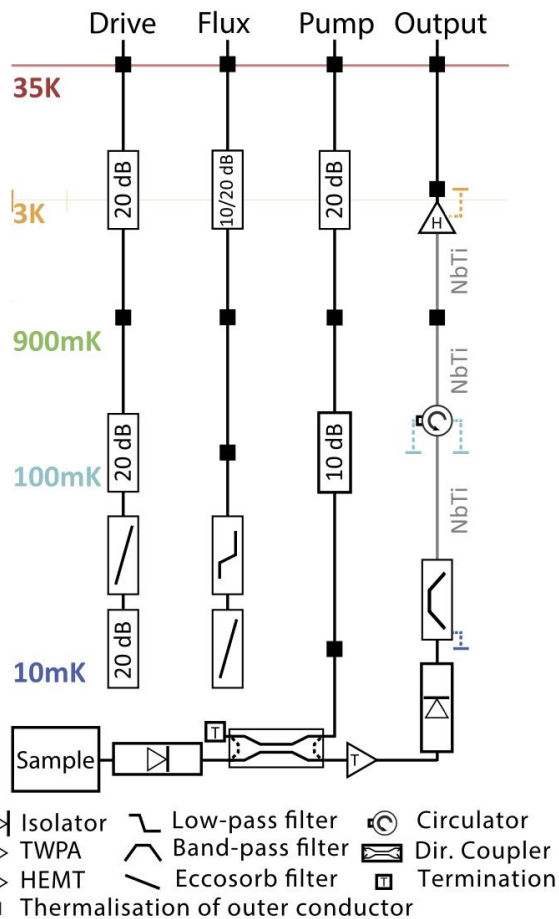
Signal, Noise, and Heat



a)



b)



Attenuator	衰減器	衰減雜訊與熱信號
Low-pass filter	低通濾波器	僅允許低頻訊號通過。~MHz
Band-pass filter	帶通濾波器	僅允許某一頻率範圍內的訊號通過。~GHz
Eccosorb (IR) filter	IR濾波器	能有效吸收高頻熱雜訊與不必要的輻射。<THz
Isolator	(雙埠) 信號隔離器	讓信號單向通過，不反射回Qubit。
Circulator	(三埠) 循環器	讓信號有方向性通過，可用作Isolator。
Dir. Coupler	(四埠) 定向耦合器	讓Pump Energy耦合到Readout Signal，有方向性，
TWPA	行波參數放大器	低溫下使用的量子放大器，具有低雜訊、寬頻帶的特性。
HEMT	高電子遷移率電晶體	常用於4K級的放大器，提供低雜訊、高增益的訊號讀取。

Attenuator and Thermal Anchor

Flexible Coax

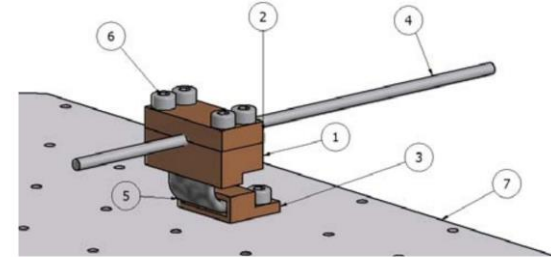
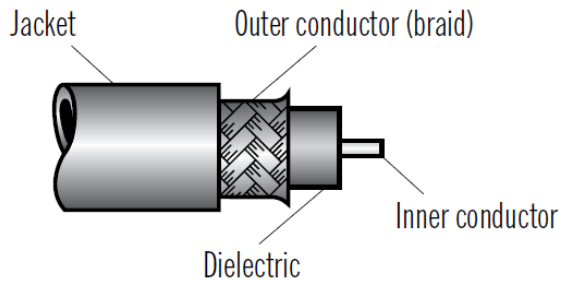


Fig. 2.8. Proposed heat sink for semi-rigid hardware; 1. Heat sink part A, 2. Heat sink part B, 3. Heat sink part C, 4. Semi-rigid coaxial cable, 5. Flexible tinned copper braid, 6. Screws, 7. Cooling base (first or second cryostat stage).

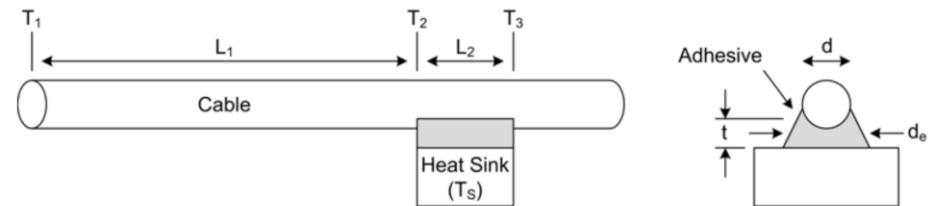
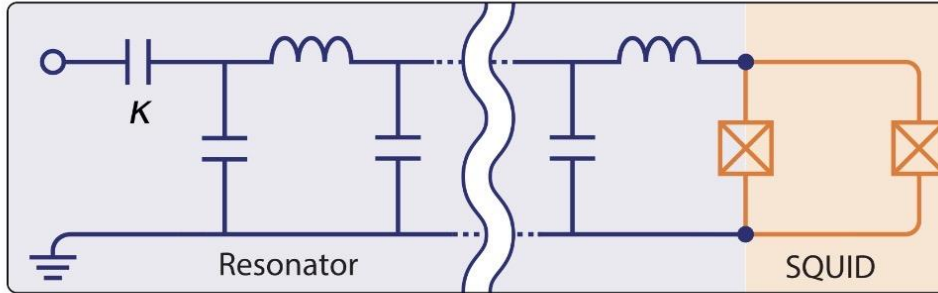


Fig. 2.7. Theoretic configuration of a heat sink.

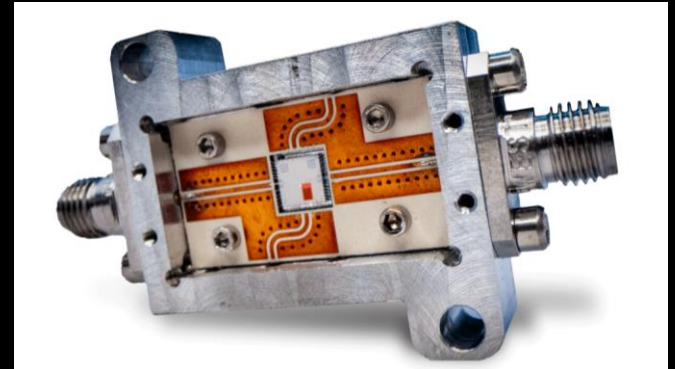
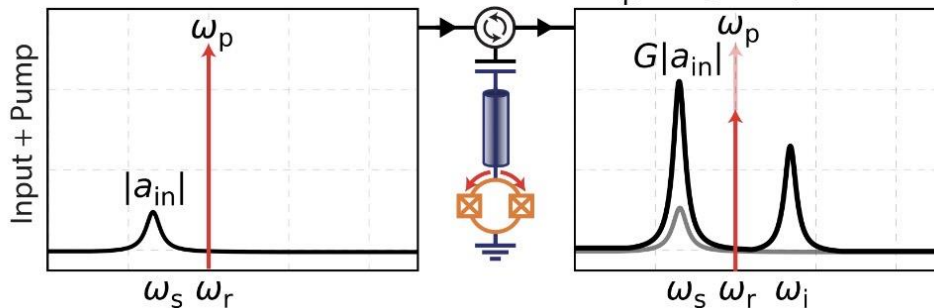
Material	T_1 (K)	T_s (K)	L_2 (mm) of heat sink for different wire sizes			
			0.21 mm ² (24 AWG)	0.032 mm ² (32 AWG)	0.013 mm ² (36 AWG)	0.005 mm ² (40 AWG)
Copper	300	80	160	57	33	19
	300	4	688	233	138	80
Phosphorus-Bronze	300	80	32	11	6	4
	300	4	38	13	7	4
Manganin	300	80	21	4	4	2
Cu ₈₆ Mn ₁₂ Ni ₂	300	4	20	7	4	2
Stainless	300	80	17	6	3	2
Steel 304	300	4	14	5	3	2

Josephson Parametric Amplifier

(a) Josephson parametric amplifier (JPA)



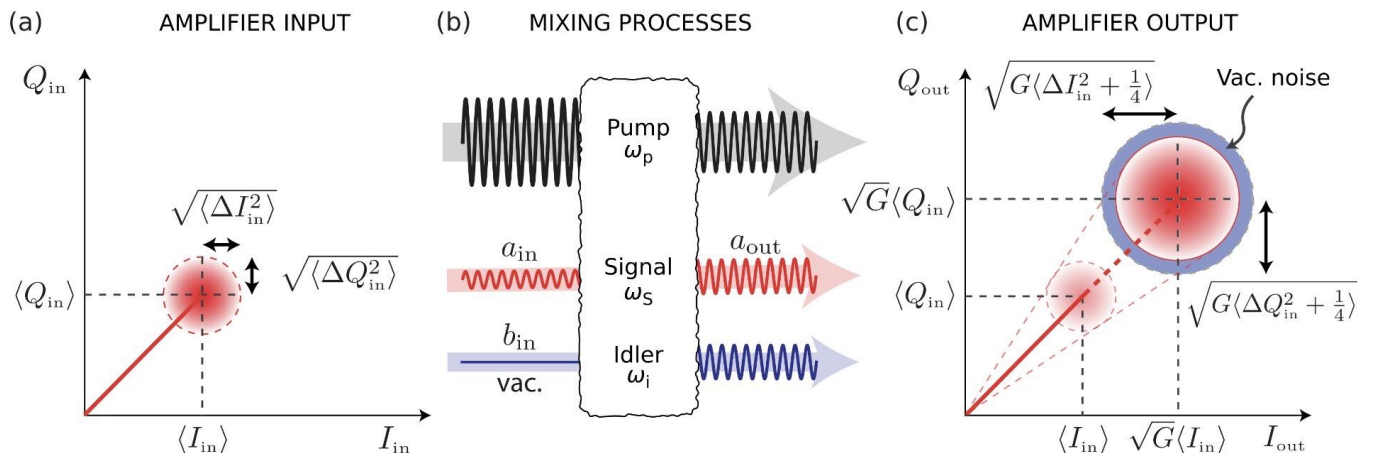
(b) Current-pumping (4-wave mixing): $2\omega_p = \omega_s + \omega_i$



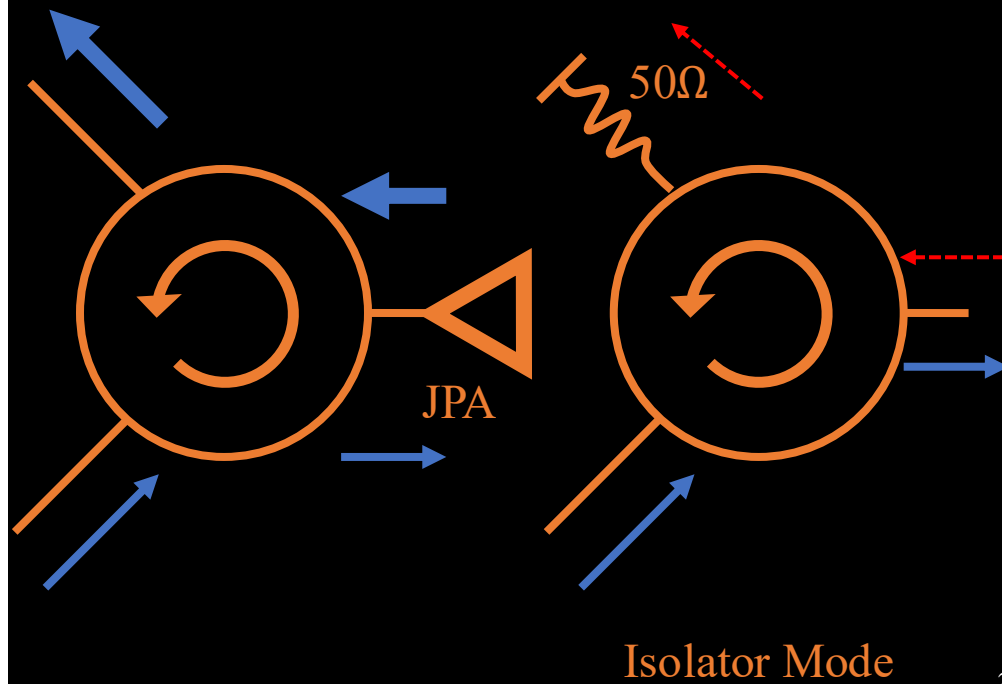
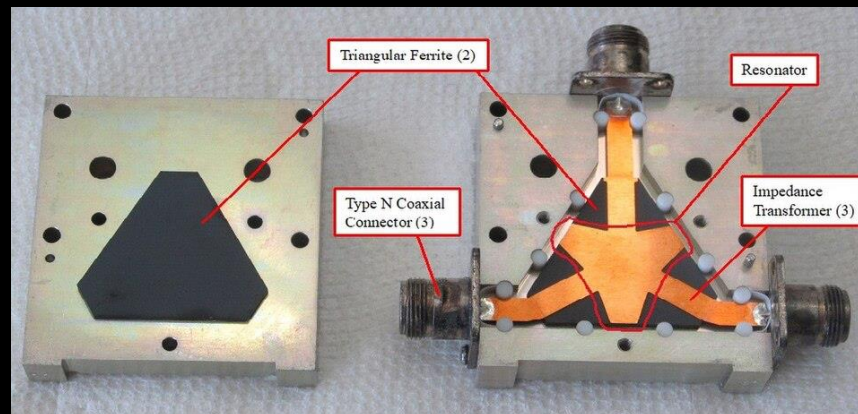
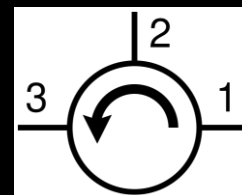
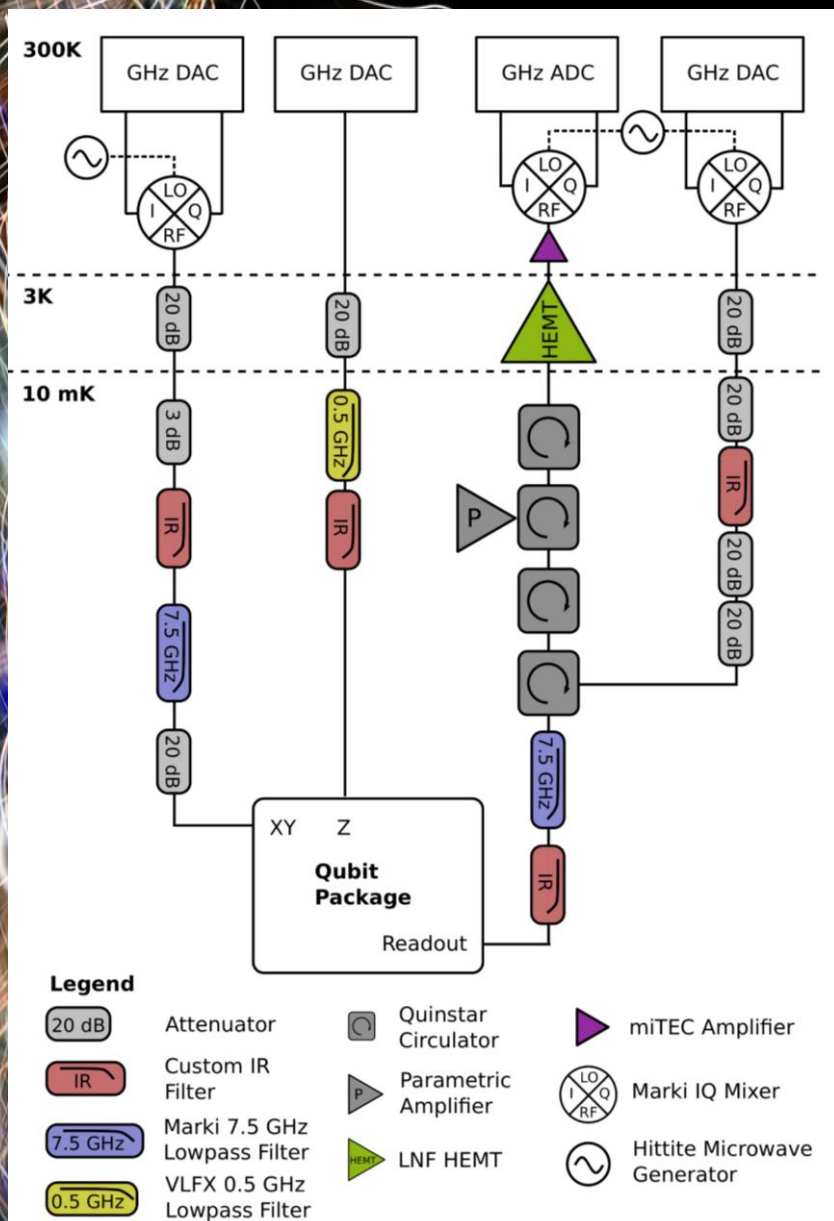
$$[a_{in}, a_{in}^\dagger] = 1$$

$$a_{in} \rightarrow a_{out} = \sqrt{G} a_{in} \quad \times$$

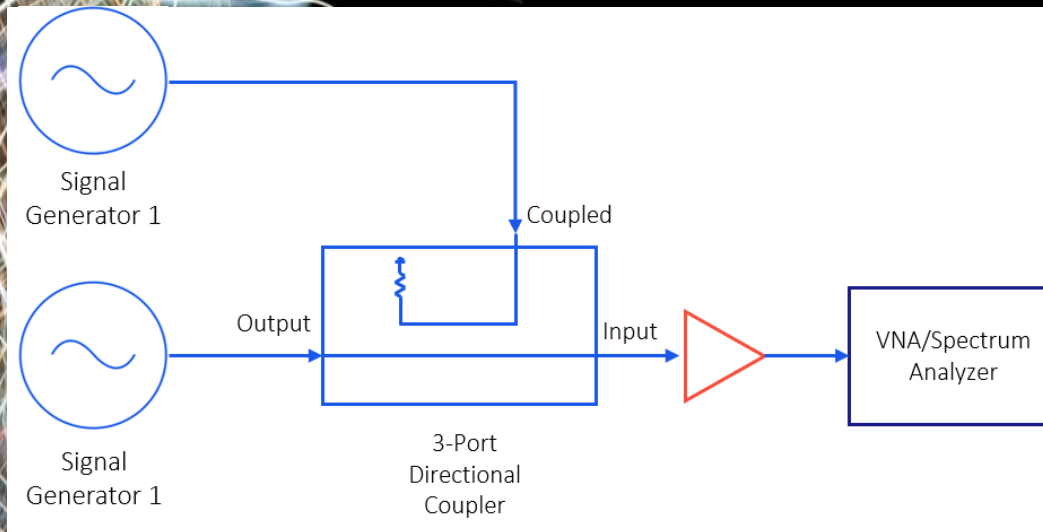
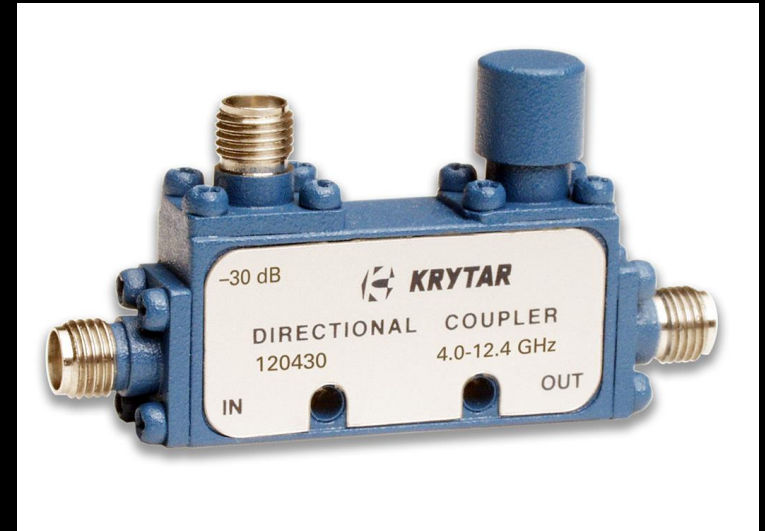
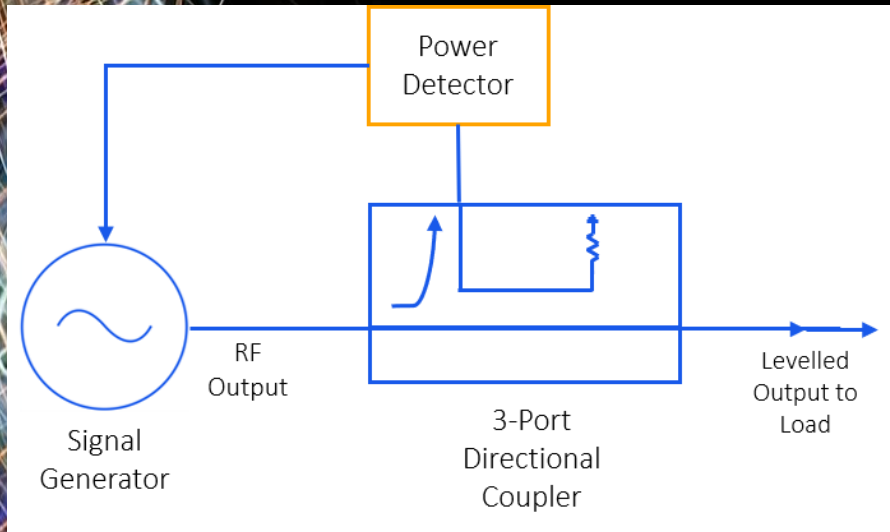
$$a_{in} \rightarrow a_{out} = \sqrt{G} a_{in} + \sqrt{G-1} b_{in}^\dagger$$



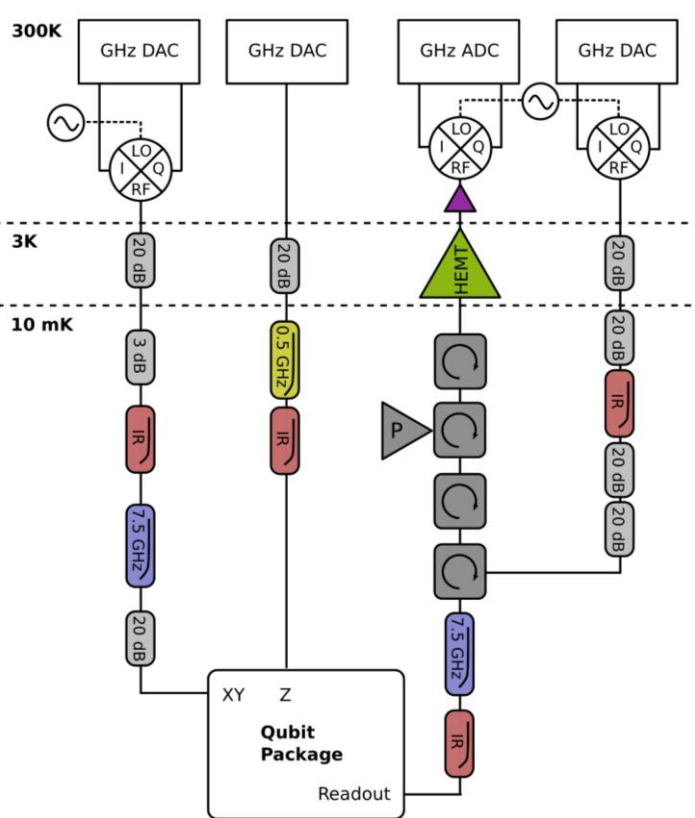
Circulator



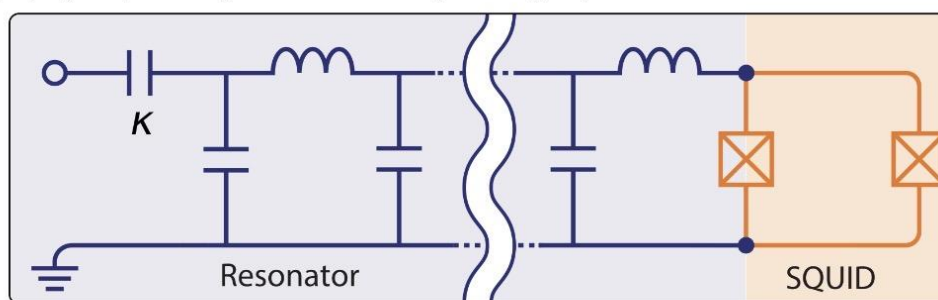
Directional Coupler



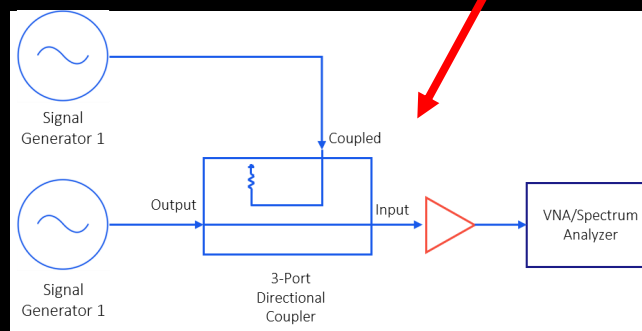
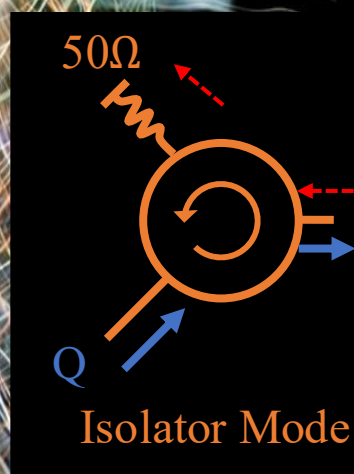
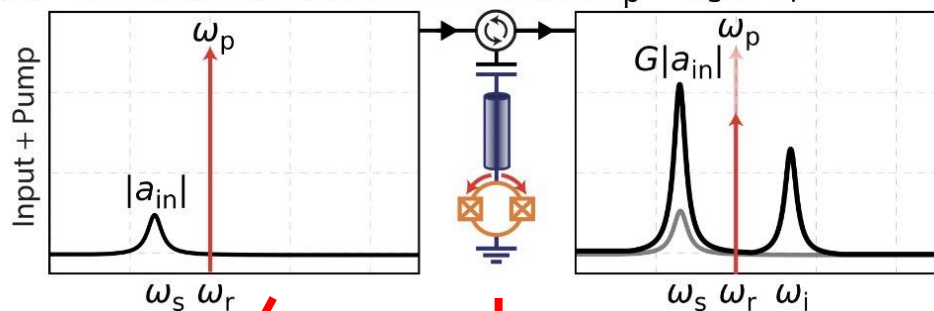
<https://blog.minicircuits.com/directional-couplers-their-operation-and-application/>
<https://krytar.com/product/couplers/directional-couplers/directional-couplers-120430/>



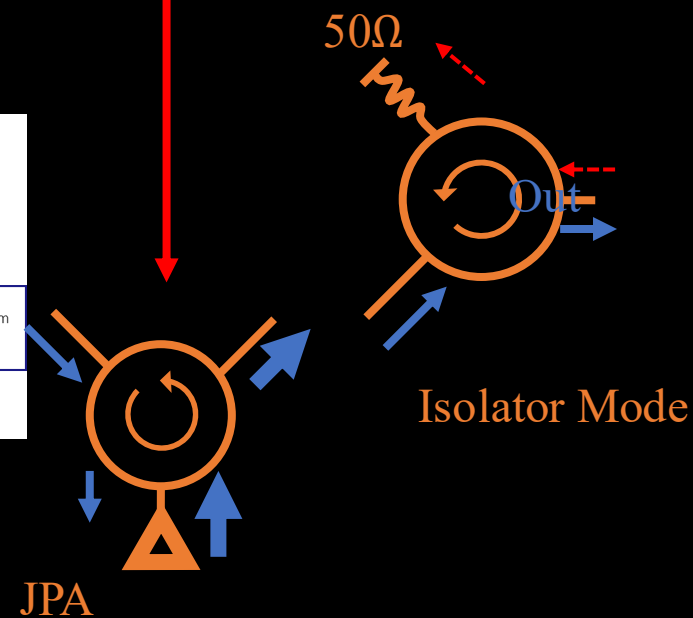
(a) Josephson parametric amplifier (JPA)

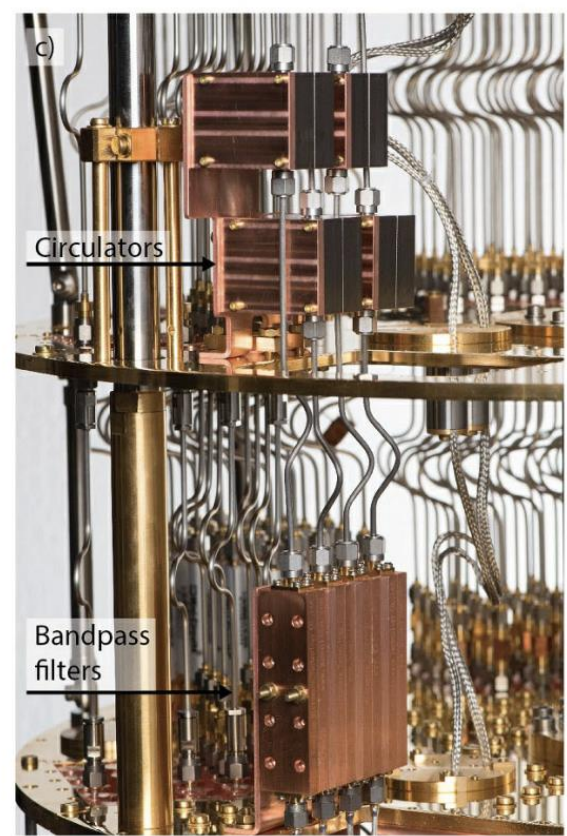
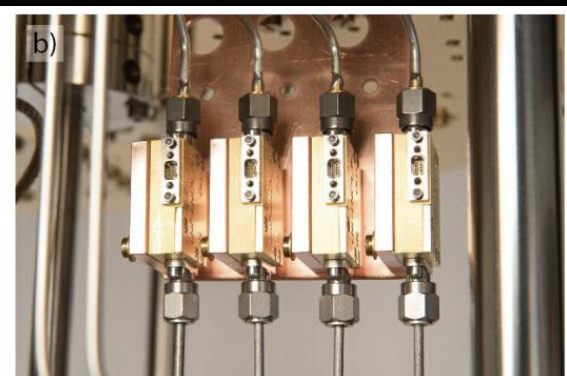
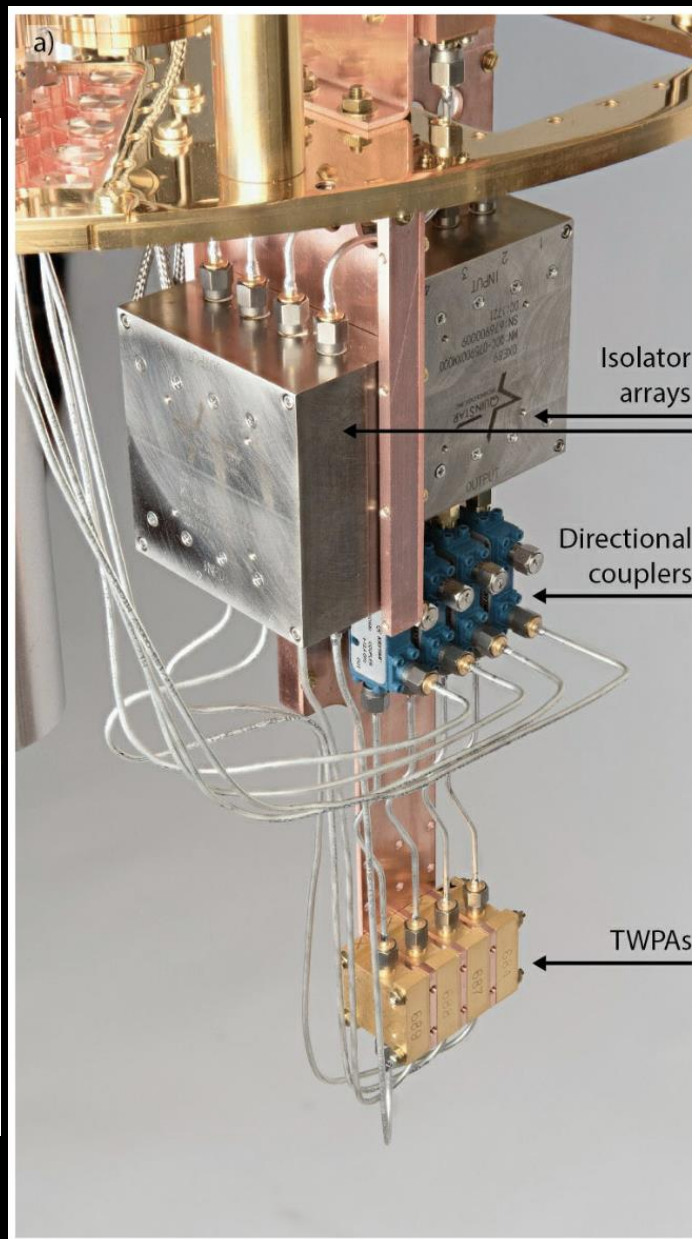
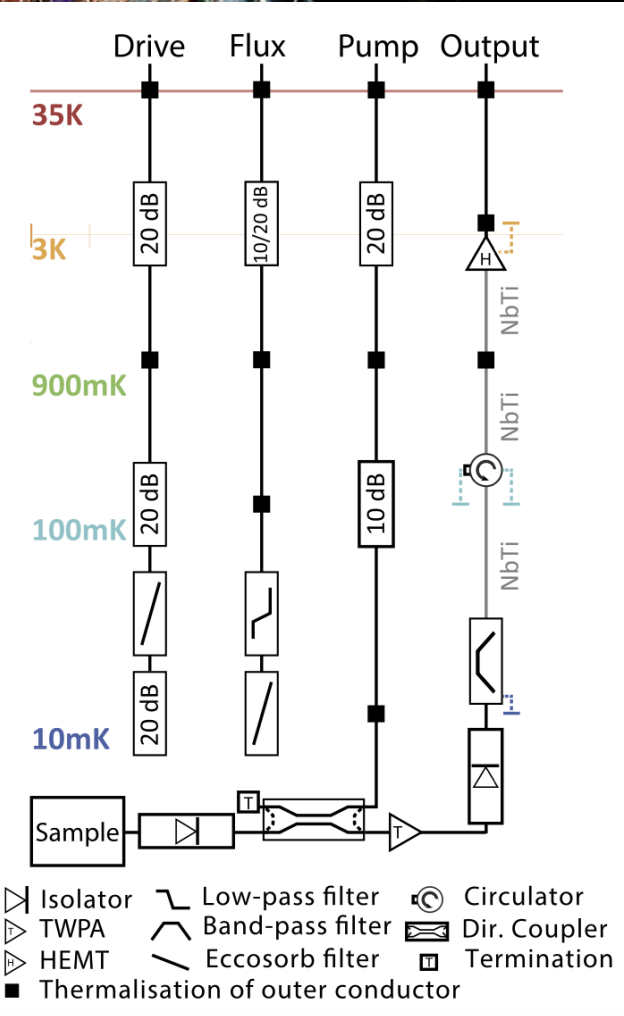


(b) Current-pumping (4-wave mixing): $2\omega_p = \omega_s + \omega_i$



Directional Coupler







Logic gates

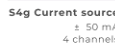
Reset

Readout

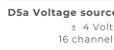
DC Sources



QSM
Source & measure
8 inputs/Outputs



S4g Current source
± 50 mA
4 channel



D5a Voltage source
± 4 Volt
16 channel



Digital



2 Input
6 Output



QTM
Timetag module
8 digital I/O



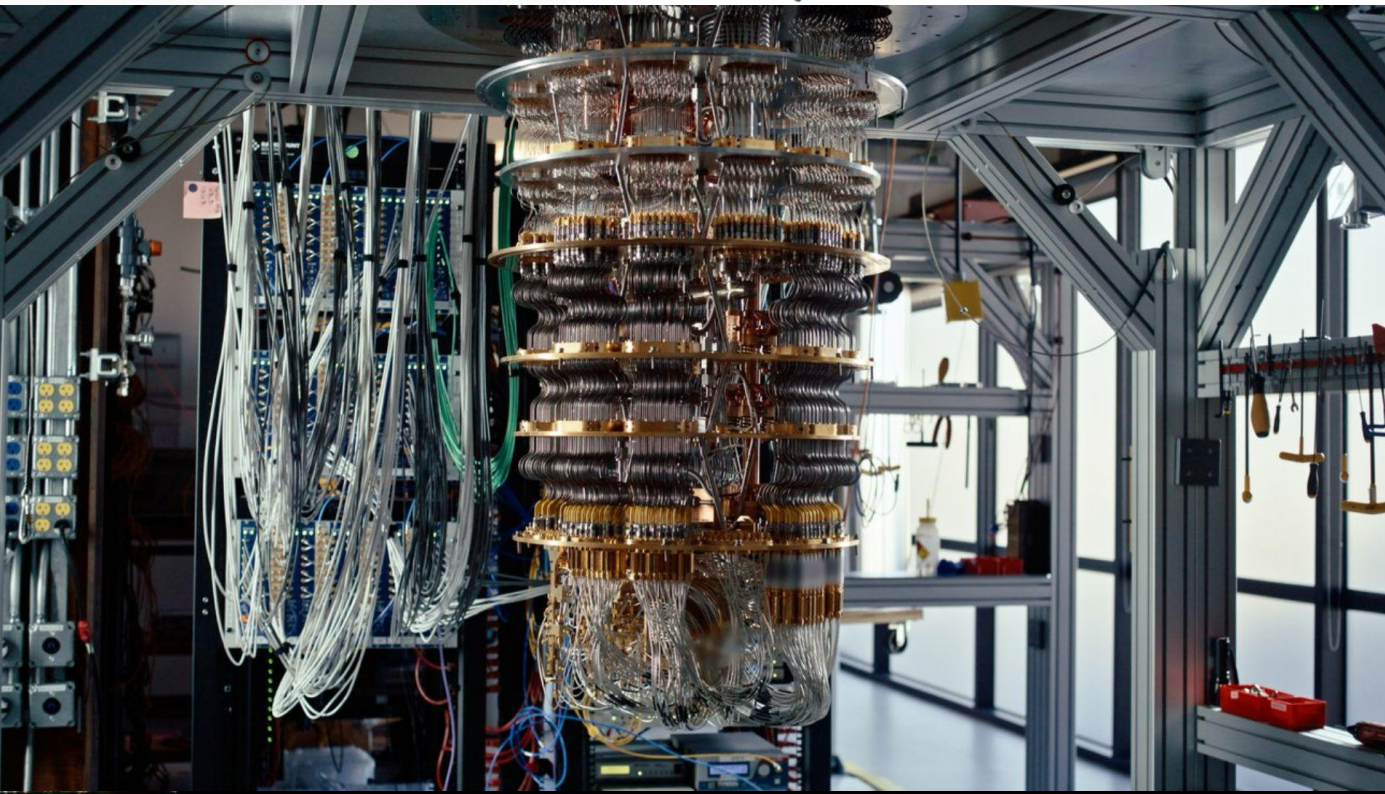
4 Outputs



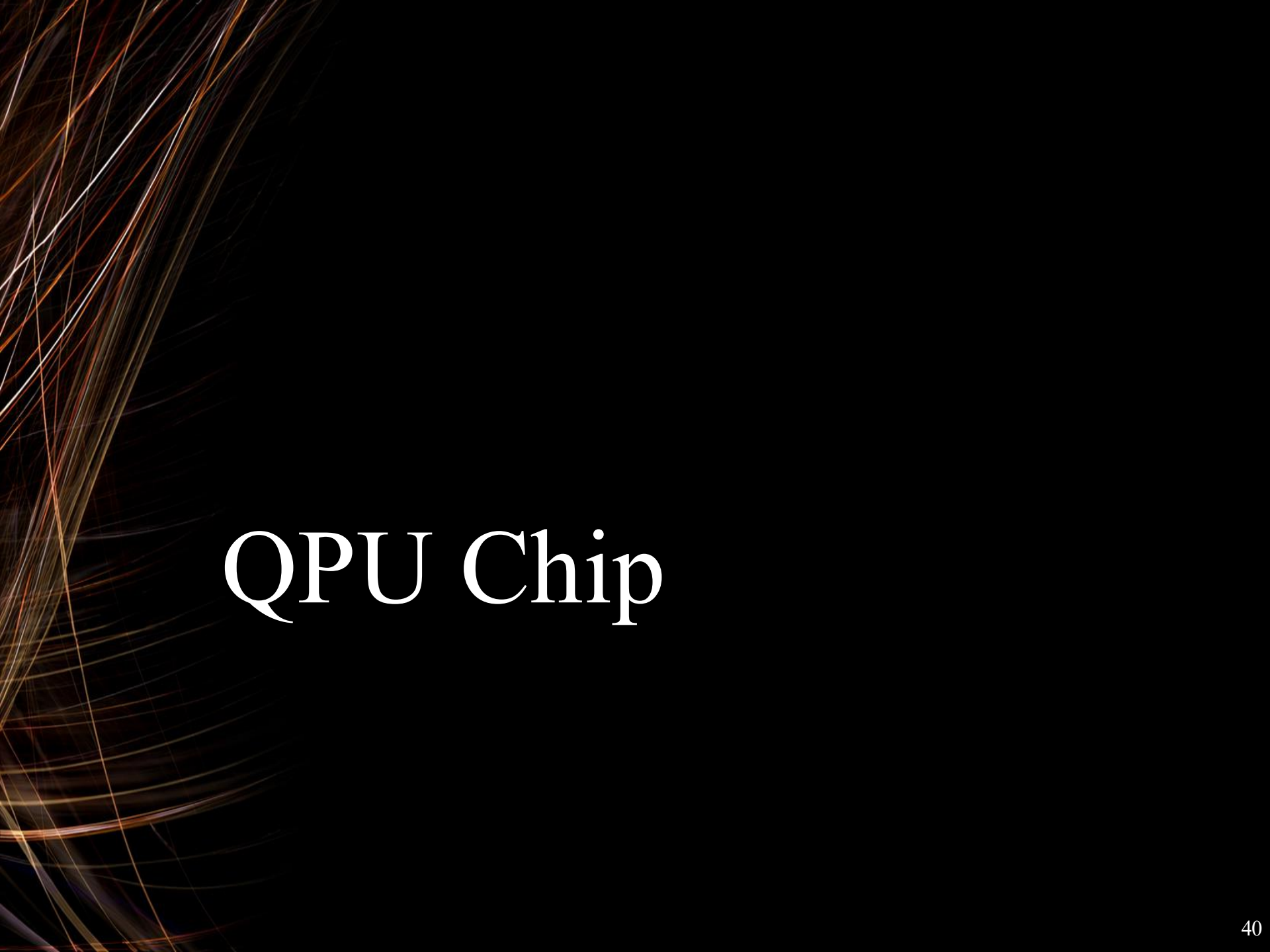
2 Inp
2 Outp



QCM
2 Outputs



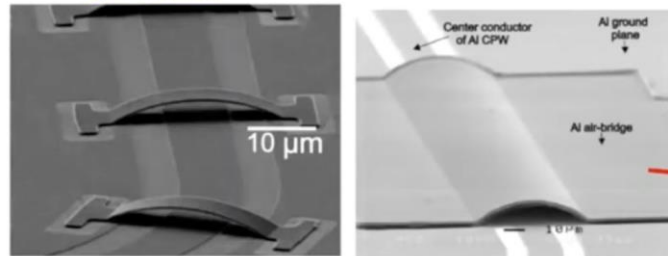
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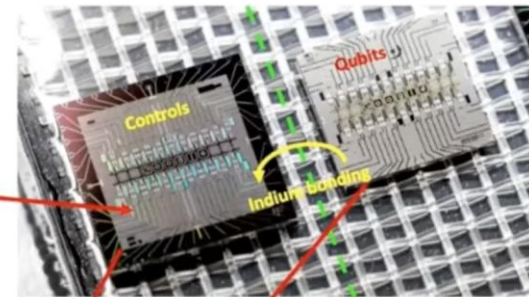
QPU Chip

晶片優化與封裝

Air Bridge 天橋結構

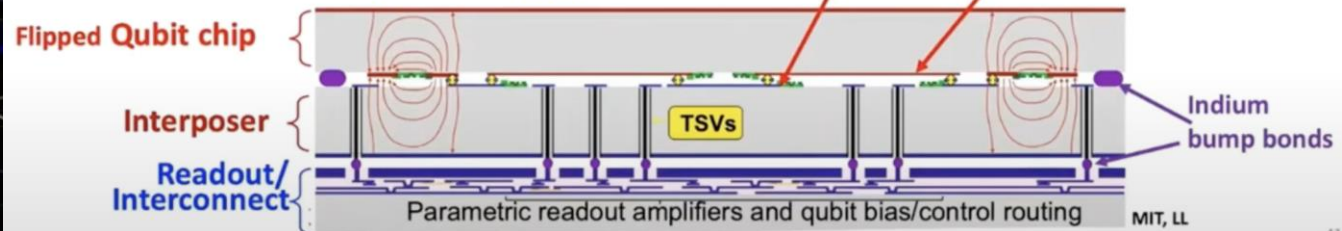


Flip chip bonding 覆晶晶片結構



Google, Rigetti

Interposer TSV 結構



Qubit plane

Transmon qubits attached to an interposer chip through bump bonds offer hardware simplicity in a scalable architecture with controllable features.

Resonator plane

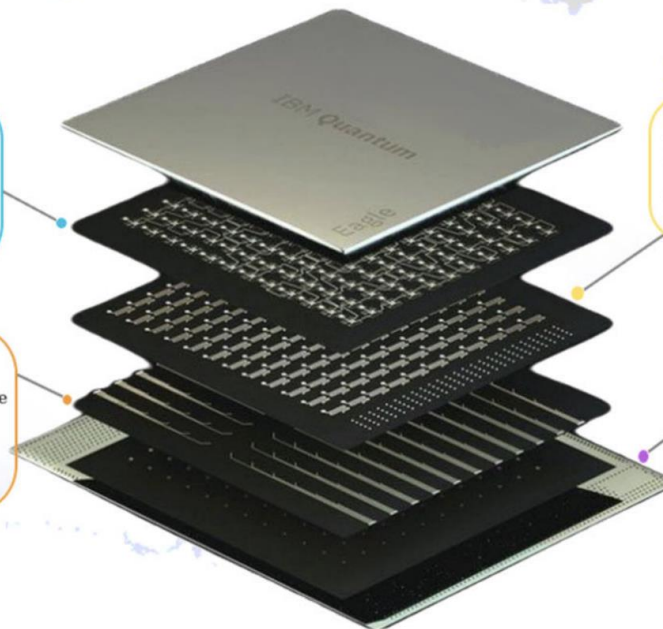
Resonators for qubit readout wired through connectors. Measured shifts in the frequency of the resonator depend on the state of the qubit.

Wiring plane

Buried wiring layer connects to the other planes through superconducting thru-substrate vias, providing the flexibility to efficiently route signals to the qubit plane with low crosstalk.

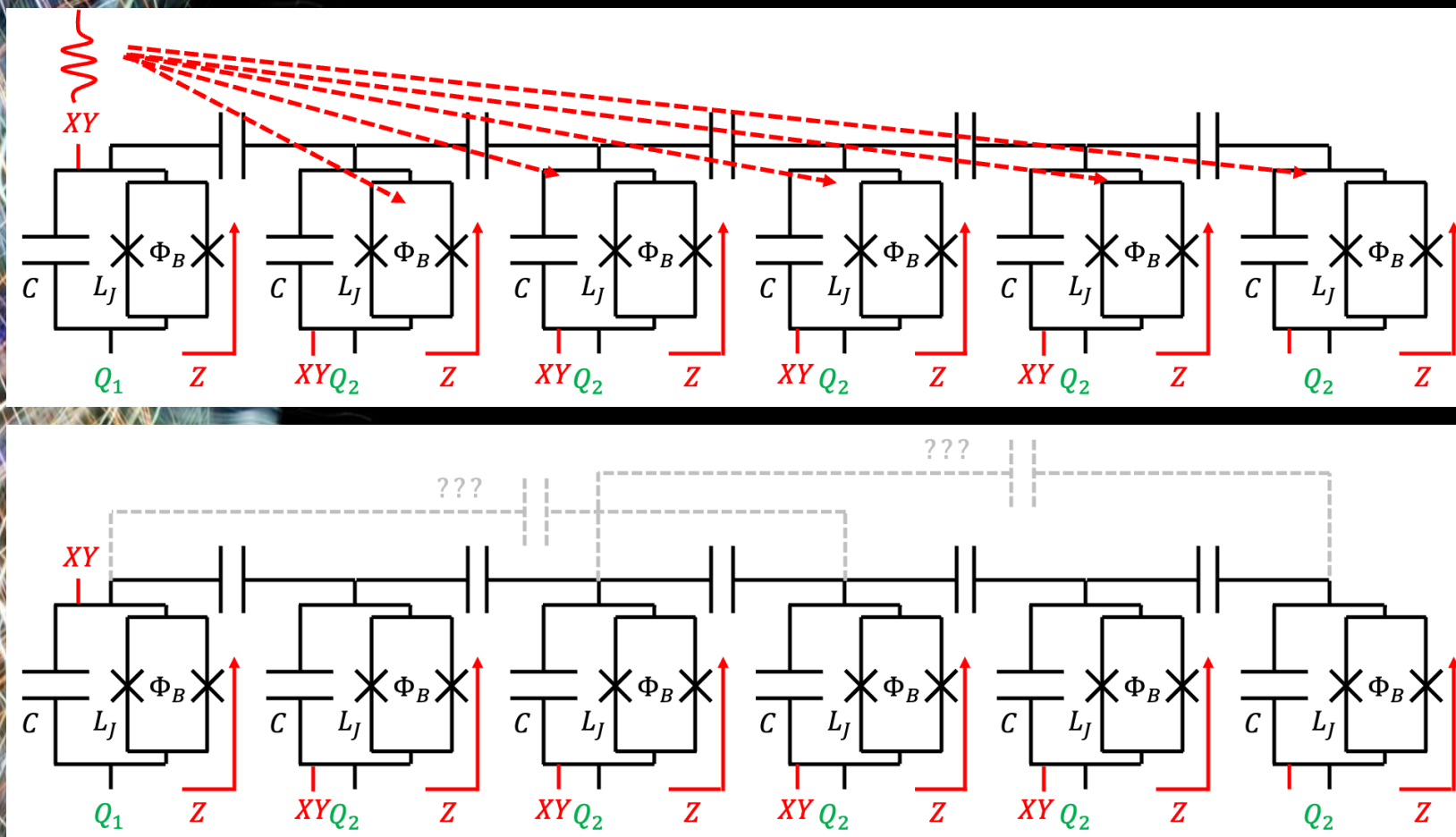
Interposer

Leverages CMOS packaging techniques, including thru-substrate vias, to exploit the third dimension to electrically connect the qubits to the other planes and deliver the signals while protecting their coherence.



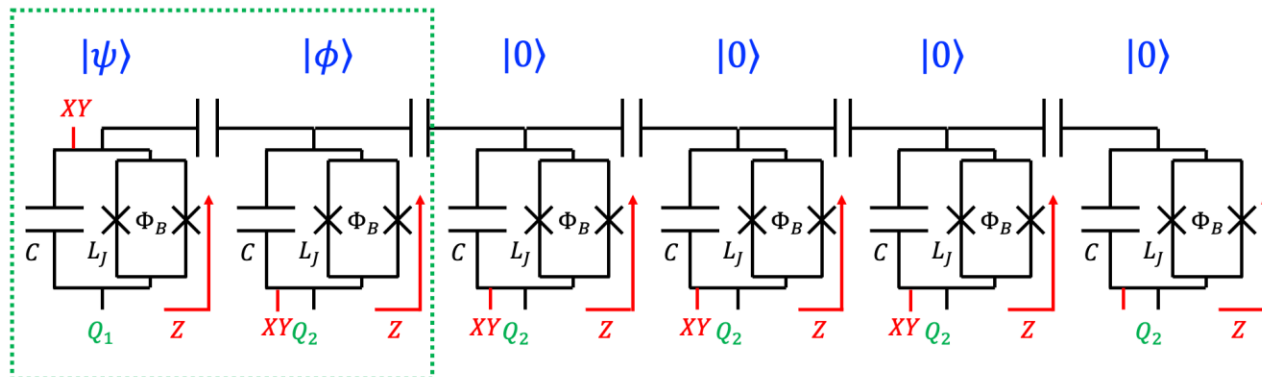
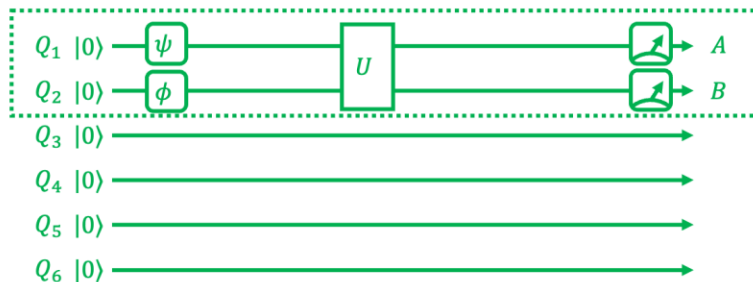
Classical Crosstalk

$$H = H_{\text{qubits}} + H_{\text{coupling}} + V_{\text{control}} + H_{\text{unexpected}}$$

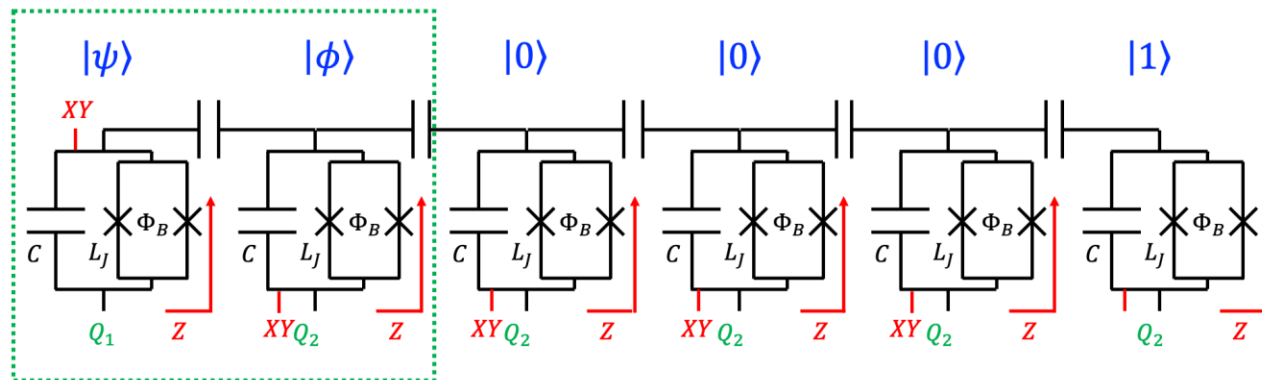
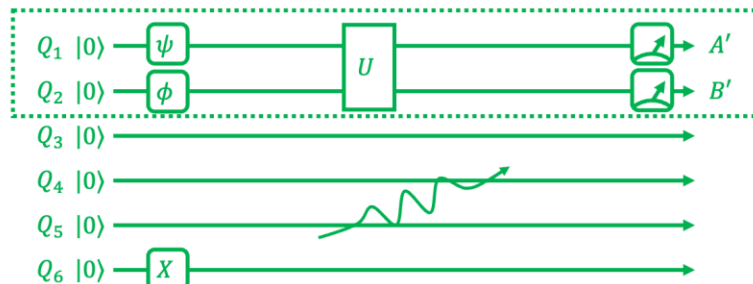


Quantum Crosstalk

(a)



(b)

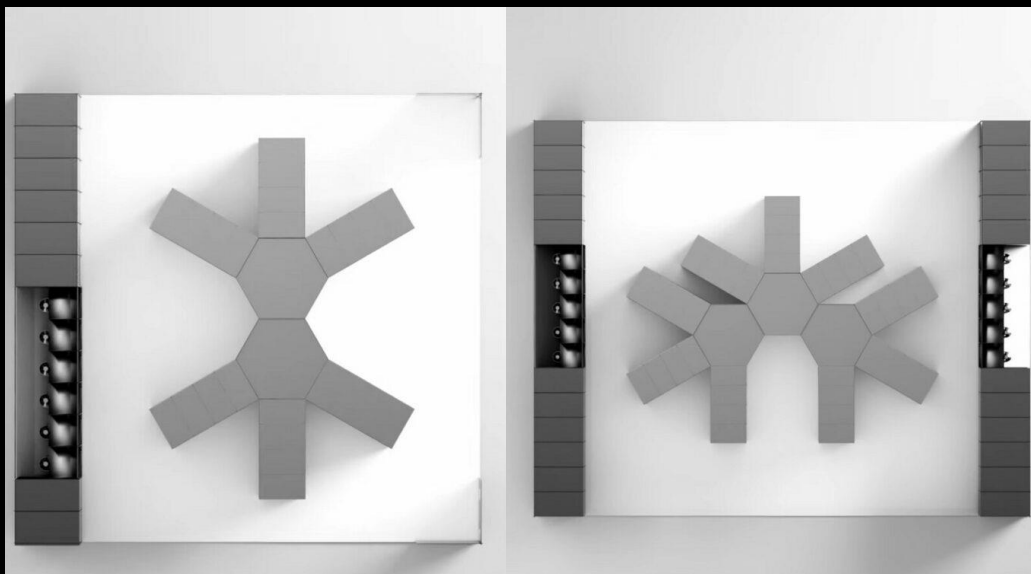


$$i\hbar\partial_t\psi = \hat{H}\psi$$



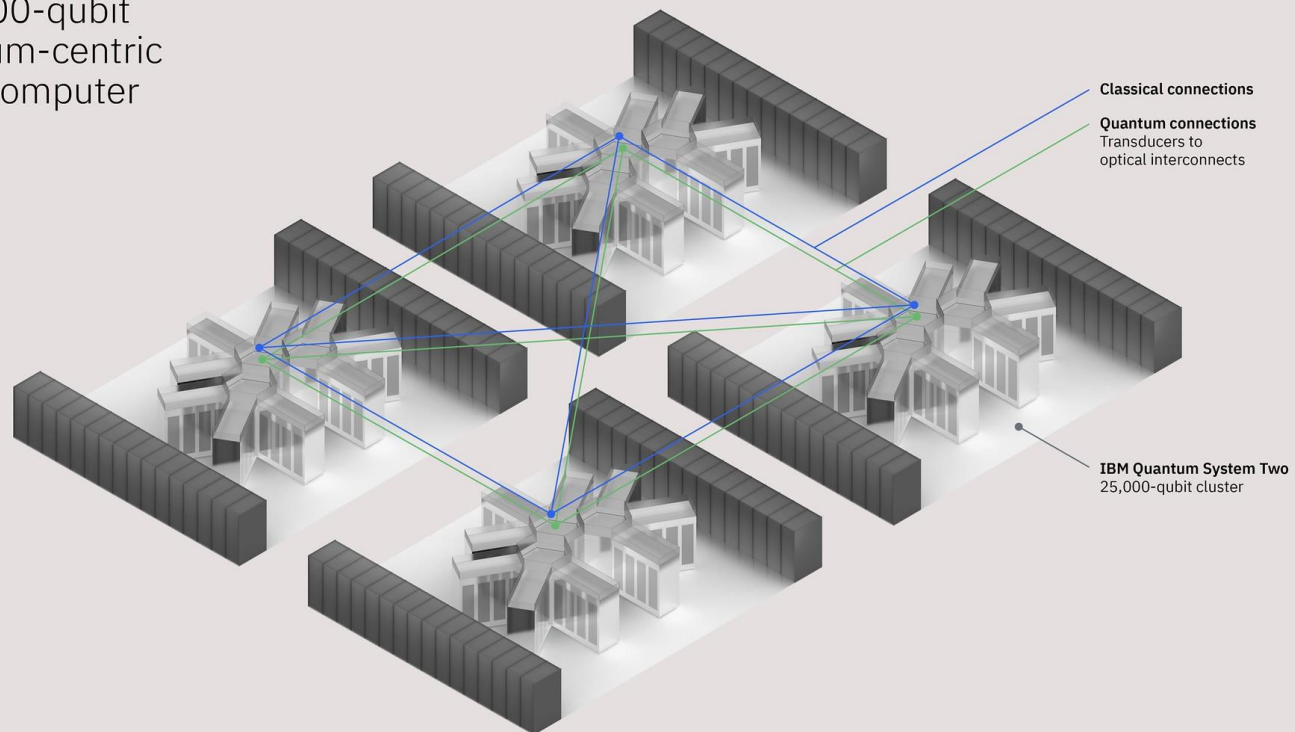
Near Future Operation

IBM HPC-QC Hybrid



100,000-qubit
quantum-centric
supercomputer

—
2033

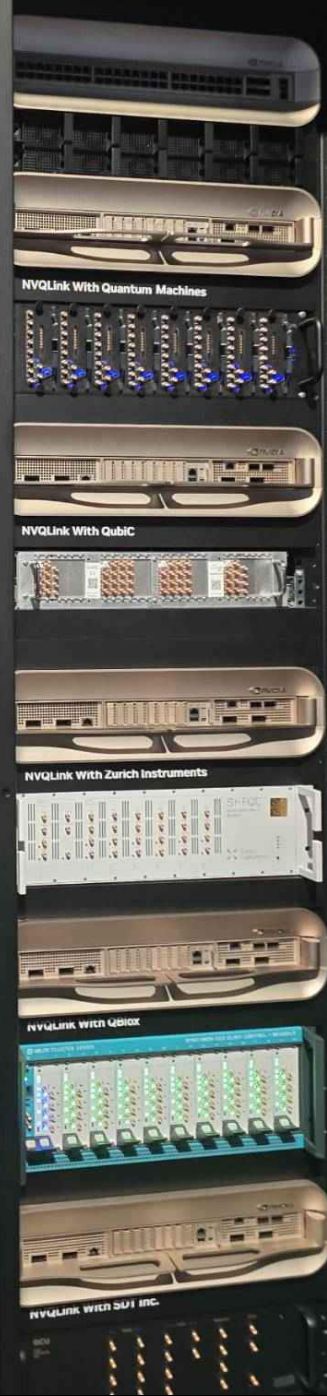


IBM Quantum

<https://newsroom.ibm.com/2023-05-21-IBM-Launches-100-Million-Partnership-with-Global-Universities-to-Develop-Novel-Technologies-Towards-a-100,000-Qubit-Quantum-Centric-Supercomputer>

Nvidia NVQLink





Quantum Machines

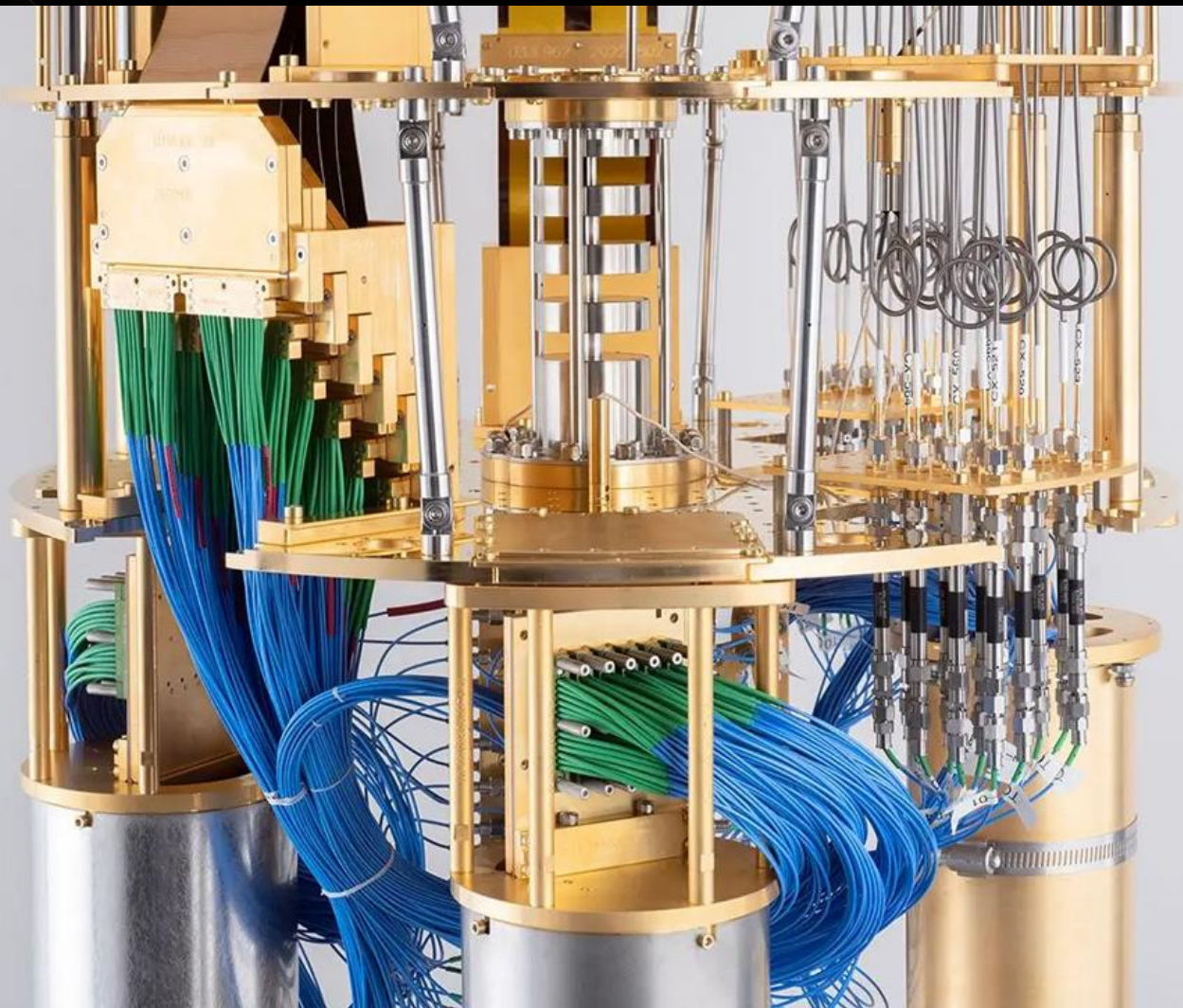
QubiC

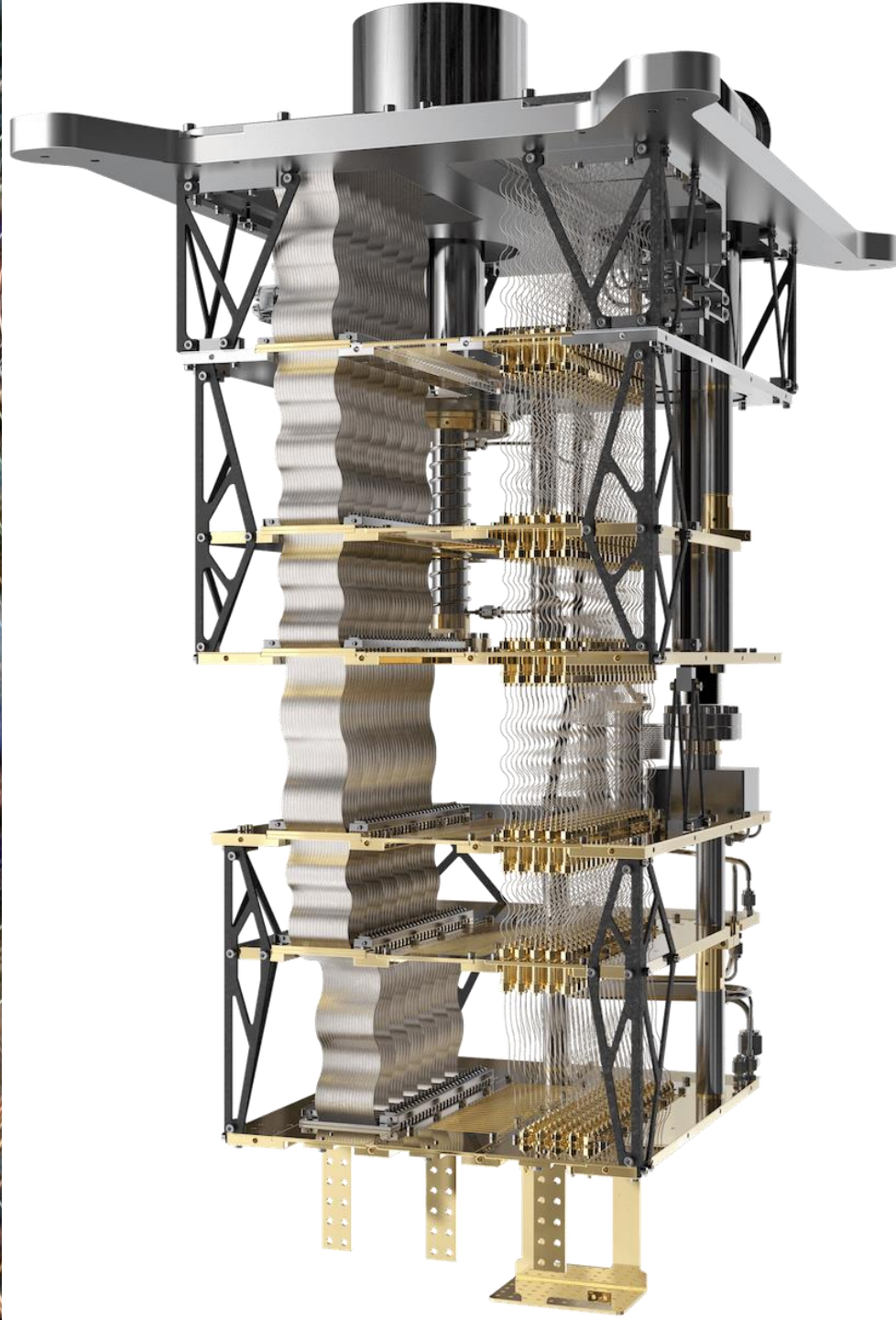
Zurich Instruments

QBlox

SDI Inc.

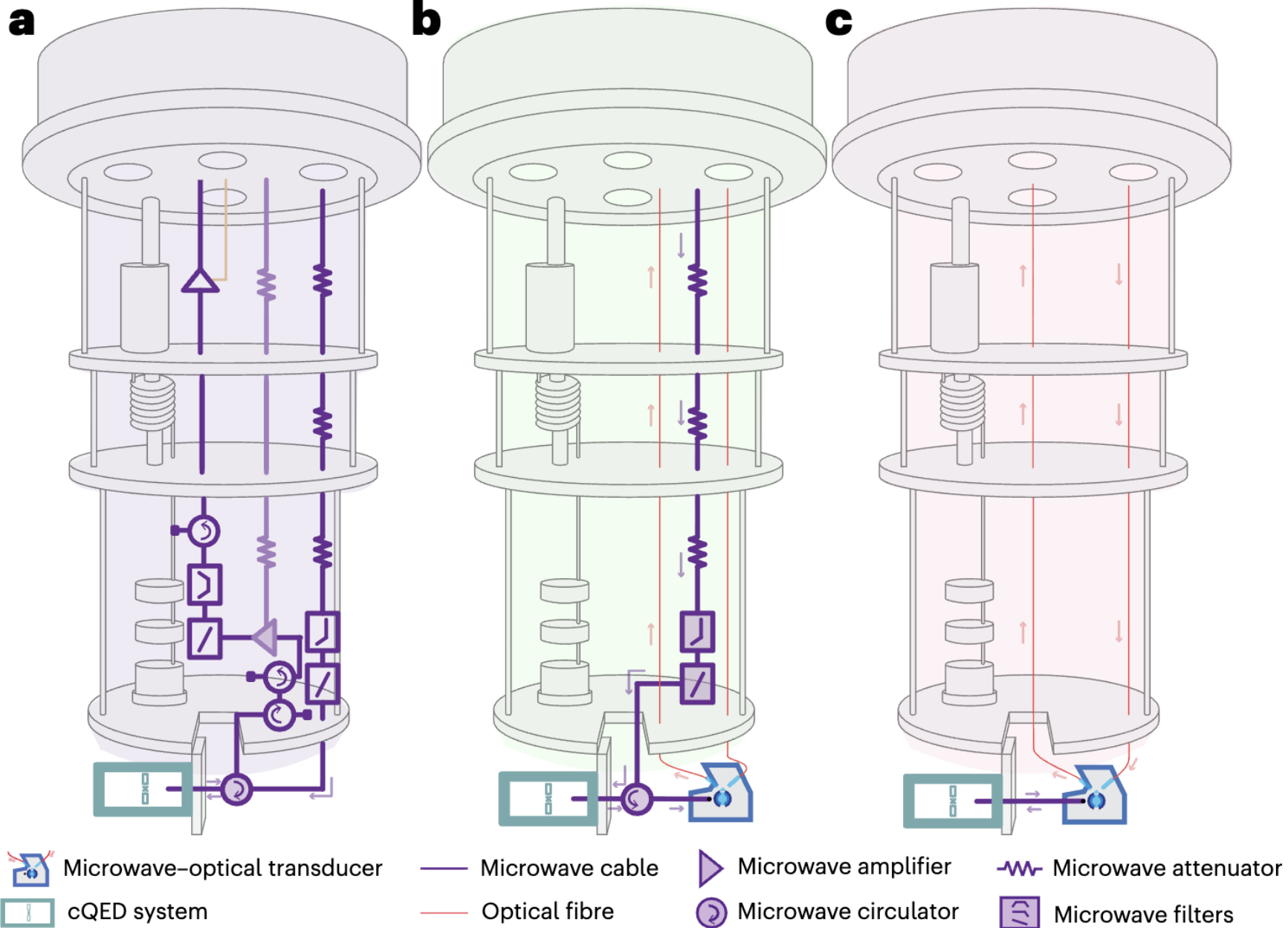
複雜走線





Maybell

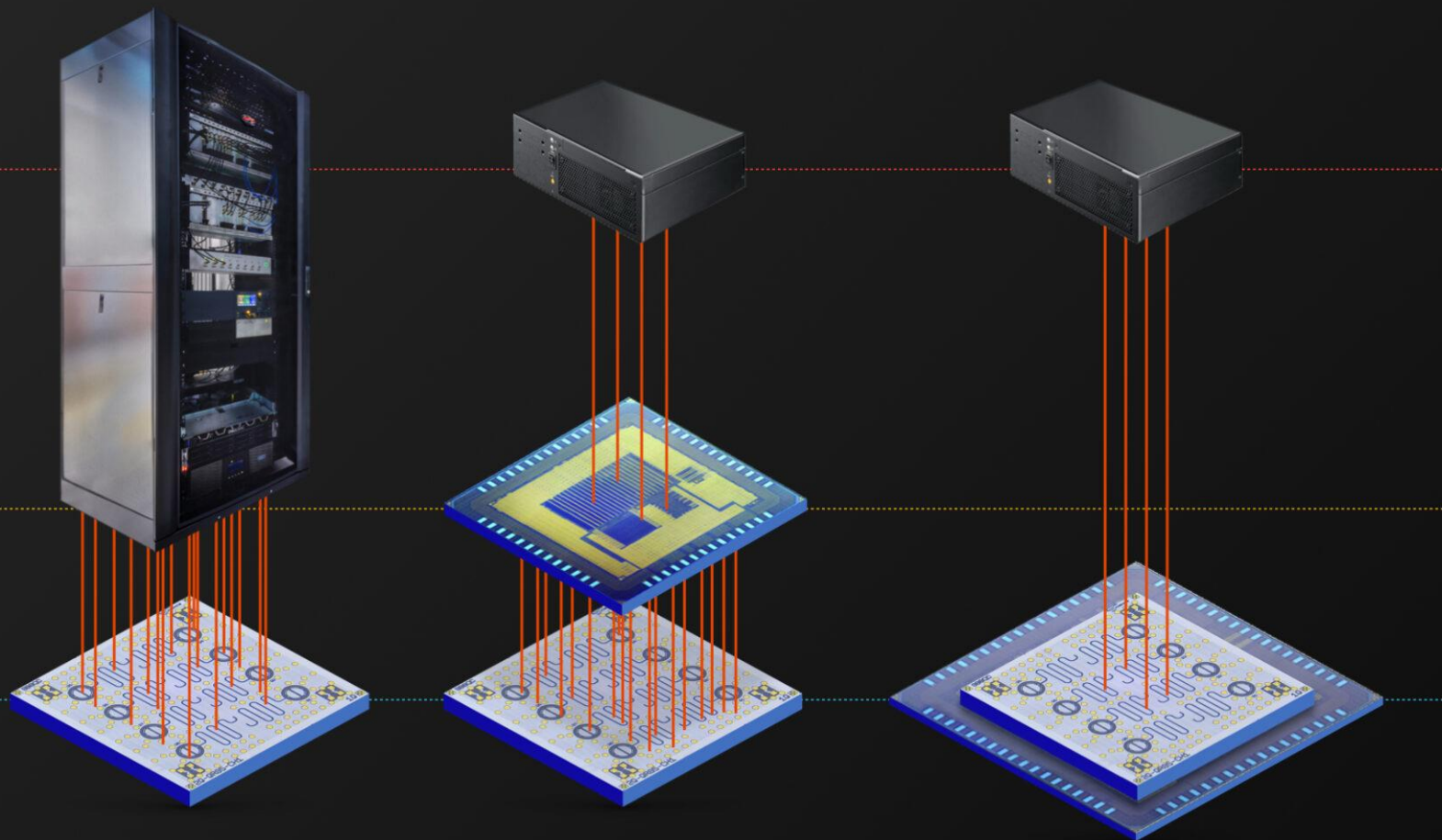




300 K

4 K

20 mK



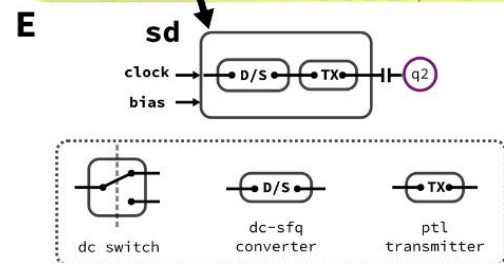
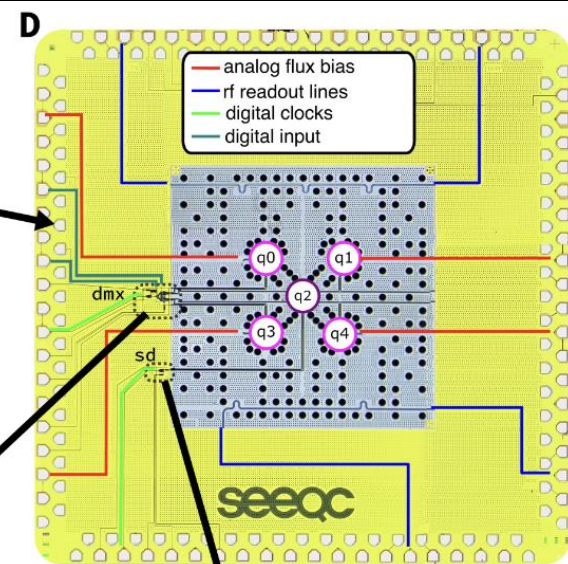
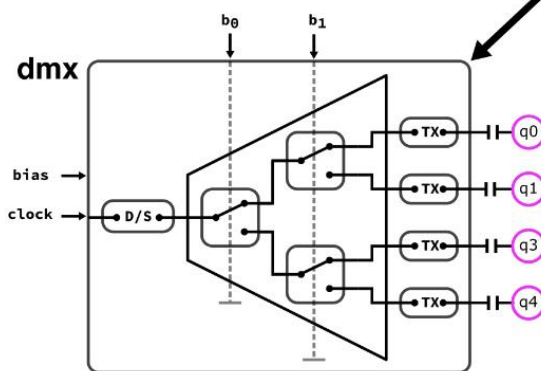
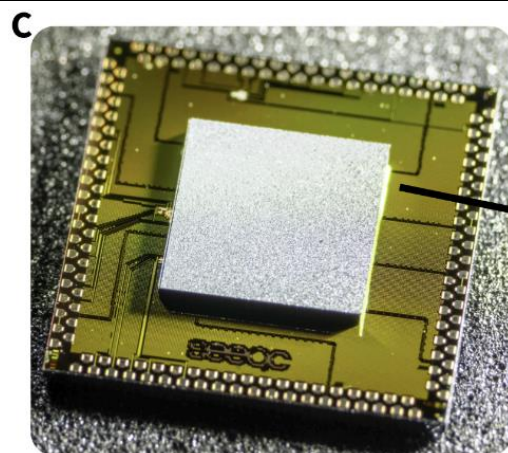
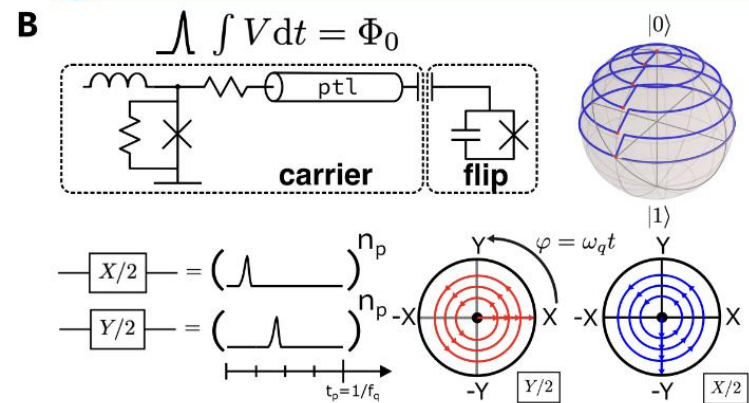
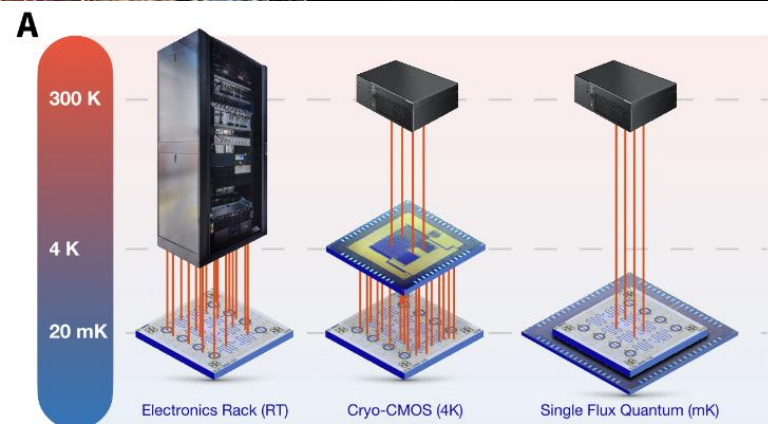
Electronics Rack (RT)

Cryo-CMOS (4K)

Single Flux Quantum (mK)

SEEQC

<https://seeqc.com/blog/innovative-chip-solutions>





Thank you